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soils4med

SOIL health monitoring and information systems FOR sustainable soil management in the MEDiterranean region

MedSIS manual

Data Managers

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DATASETS | MAPS | DOCUMENTS | GEOSTORIES | DASHBOARDS

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1. Overview

This manual has been developed as a reference guide to assist Med-SIS STAFF and Data Managers in navigating and using the Med Soil Information System (Med SIS).

The SIS is essentially a **Web GIS Catalogue** of resources related to soil data. It is based on Geonode (https://geonode.org/), whose resources include datasets, maps, documents, geostories and dashboards.

In addition to the standard catalogue functionalities for managing **soil data**, MED-SIS includes an additional tool, known as the **Back Office**

MED-SIS staff and data managers use the Back Office to create and configure catalogue resources and to manage them through the Web GIS catalogue.

This document is divided in three main sections: “Web GIS Catalogue”, “Back Office”, “Admin panel”

Further details on Web GIS Catalogue functionalities can be found on the official GeoNode documentation page at https://docs.geonode.org/en/latest by selecting the “User Guide” menu

2. Web GIS Catalogue

Within the SIS, only a limited subset of content is publicly accessible. Registered generic users therefore have read-only access and can interact with resources only in accordance with the permissions explicitly defined by the data manager responsible for each dataset.

The data manager is responsible for data ingestion into the Med-SIS and **defines the access control policies** applied to each dataset, **based on the intended users and their roles**. The data manager cant add datasets and documents, create maps, dashboards and geostories, and edit and share resources with other users.

Med-SIS homepage

The MEd-SIS system has a moderate user registration by default; this means that the users must be validated by the staff after the registration (Figure 1) .

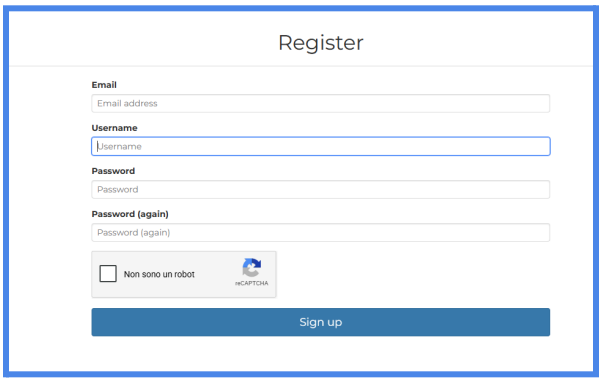


Figure. 1 Registration page

Therefore, registration requires an exchange of emails between the system administrator and the user. The system administrator's contact details are available on the "Data Policy" page, accessible via the link at the bottom of the homepage.

Once registered and approved, users can access the catalogue .using the “Sign in” link in the top of the home page

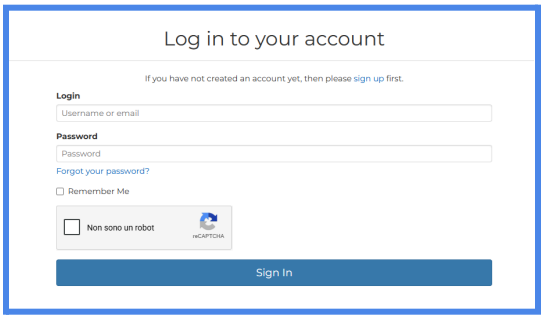


Figure. 2 Sign in page

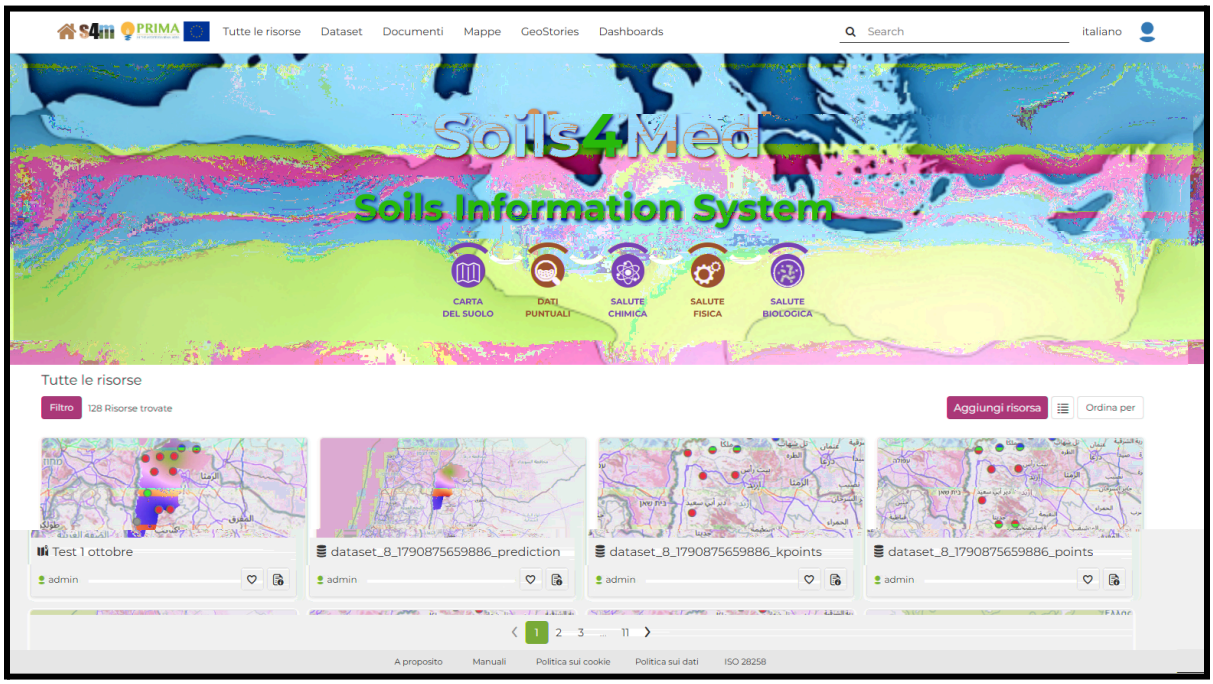


Figure. 3 Med-SIS homepage after the login

As shown in Figure 3 the system's main page displays the project logo in the upper-left corner. Clicking the logo redirects the user to the homepage .



Figure 4. Bottom bar

The bottom bar of all pages shows some link:

- **About**: to display site description
- **Manuals**: to display help page
- **Cookies** Policy: to display cookies policy
- **Data Policy:** to display data policy of the site
- **ISO 28258**: to display the help page on access Soils4Med data in ISO 28258 standard

In the upper-right corner, there is the language selection menu and the user menu which provides access to user-related actions, as illustrated in Figure 5.

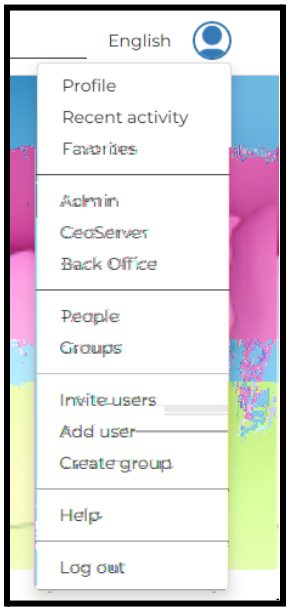


Figure 5

- ***Profile***: displays the user's profile
- ***Recent activit*** : displays the user's recent activities
- **Favorites**: displays the list of the favorites datasets
- **Admin**: opens the admin panel if the user is a member of the staff
- **Geoserver**: opens the Geoserver application
- **Back Office**: open the back office if the user is a member of the staff
- **People, Groups, Invite user, Add User, Create Group**: Users and groups management - only if the user is a member of the staff
- **Help**: provides quick help on performing actions in the catalogue
- **Log out**: logs the user out of the system.

In the top area of the MED-SIS homepage, there are menus providing direct access to the available resources, organized by category (Datasets, Maps, Documents, Geostories, and Dashboards) or collectively under All Resources.

The same resources can also be accessed by scrolling down the page.

In the center of the page, there are quick links providing direct access to the following resources:

- **Soil Maps** : Vector / Raster datasets containing Pedological Classification
- **Soil Point Data** : Point vector datasets containing soil profile and samples data
- **Chemical Health** : Vector / Raster datasets containing soil indicators describing chemical health
- **Physical Health**: Vector / Raster datasets containing soil indicators describing physical health
- **Biological Health**: Vector / Raster datasets containing soil indicators describing biological health
- **Other Maps**: Vector / Raster datasets containing generic gis data
- **Help Manuals**: MED-SIS manuals in pdf format like this data



Figure. 6 Resources' List and Filters

Once the list of available resources is displayed (Figure. 6):, resources can be filtered using one or more of the following filters:

- **Featured**: resources explicitly marked by data managers or administrators as highlighted or recommended
- **Remote**: resources served from an external service, such as a Web Map Service (WMS), an ArcGIS REST service, or another GeoNode instance
- **Dataset**: filters datasets by type including Vector, Raster, Time series, and 3D Tiles
- **Documents**: filters the list to show available documents
- **Maps**: filters the list to show available maps including maps s created using the h Geonode Map Viewer, maps created through MapStore-based viewer configuration files, maps created via API, or scripts, and maps imported from other Geonode instances
- **MapViewers**: filters the list to show only maps created using the Geonode Map Viewer
- **GeoStories**: filters the list to show available Geostories
- **Dashboards**: filters the list to show available Dashboards
- **Category**: filters resources based on the value assigned to this t metadata field. Typical categories, often based on INSPIRE / ISO topic categories used in geospatial metadata standards, include Boundaries, Environment, Transportation, Economy, Health, Society, and Utilities / Communication. MED-SIS also includes ***Point Soil Data Chemical Health Ph sical Health*** and ***Biological Health***.
- **Keywords**: filters resources based on the keywords assigned in the metadata, typically free text tags. Several Med-SIS keywords have been added such as "Pedological map"
- **Regions**: filters resources based on the geographical region specified in the metadata.This is different from the bounding box, which is automatically derived from the geographical extent of the data.

- **Owner**: filters resources based on the user who created or owns them
- **Group**: filters resources associated with a specific GeoNode group
- **Date from**: filters resources based on their creation or upload date, showing only resources added on or after the specified date
- **Date to**: filters resources based on their creation or upload date, showing only resources added on or before the specified date e
- **Extent**: filters resources based on their geographical extent, including resources whose geographical area intersects or is contained within the selected extent

User Profile

Once registered, the SIS administrator assigns new members the Data Manager role in accordance with the system's role-based access control policies.

Communication among the different actors within the SIS can take place through the platform's internal messaging system or via private email.

By selecting the "Profile" option from the user menu, the data manager is directed to the profile management interface, where they can:

- configure personal information by clicking the *Edit Profile* button. Users can update their profile by adding a profile photo, specifying their organization, writing a description, indicating their position, and setting their location
- manage email settings by clicking the *Associates emails* button
- manage credentials such as the password, by clicking the *Set/Change password* button
- access a summary of their activities in the SIS by clicking the *My Activities* button, which displays the user's latest actions within the system
- configure notification settings, including how and when to receive notifications and for which events, concerning activities on the platform by other users or administrators, by clicking the *Notification* button
- creating announcements

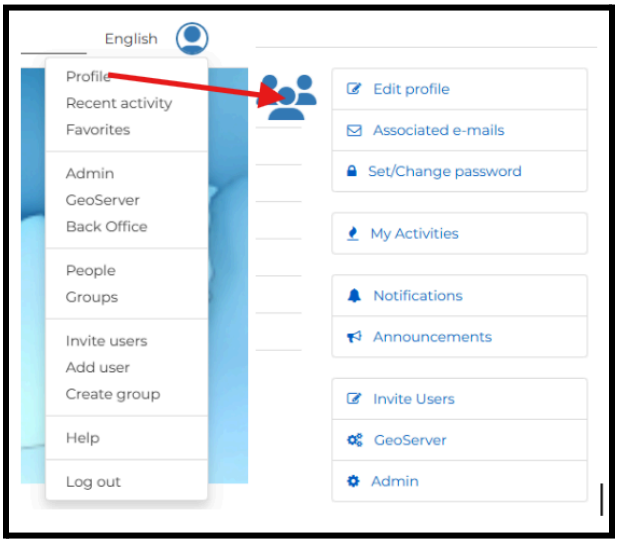


Figure 7 User profile menus

The **Recent Activity** option provides a detailed overview of the actions performed within the SIS by active users. It allows users to view the type of activities performed and identify the users who performed them.

The **Favorites** option provides access to the list of resources marked as *favorites,* allowing users to easily retrieve them without having to search for them in the catalogue.

The **Inbox** option is part of the Geonode internal messaging system and is designed to facilitate communication between registered users and group members. It acts as a hub for receiving, reading, and replying to messages within the SIS platform.

The other options in the menu provide access to administrative tools such as **Admin** and **Geoserver**, the **Back Office** for uploading and managing soil data, the **Help** section, which provides guidance on using GeoNode functionalities and the Log **out** button.

The *Admin Panel* and *Back Office* are described in the respective sections of this manual. Geoserver, which is responsible for publishing and serving the spatial data of the SIS platform through standard OGC web services such as WMS, WFS, and WCS, should not be used by data managers and is intended for use by the system administrators only.

Resources

The data manager can manage all existing catalogue resources, which are organized into the following categories:

Dataset

Dataset are geospatial resources that can be stored, managed, visualized, and shared through the platform. A dataset typically represents a spatial layer.
The Dataset view has several menus and options for managing and accessing the resource.



Figure 8 Resource menu

by clicking on the **Open Property button**, the user can view the resource's general information, geographic location, alphanumeric attributes, and linked resources that use or are used by the dataset

By clicking the **Resource menu,** users with staff permissions can save or delete the resource.

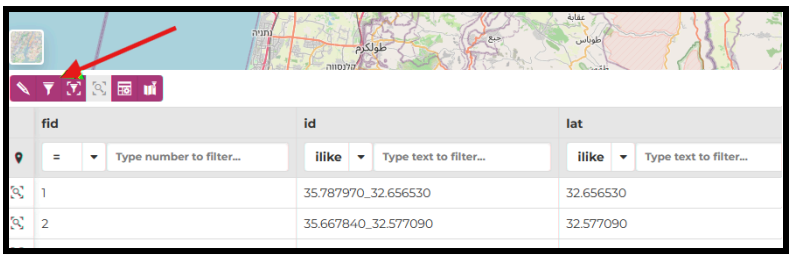


Figure 9 View Data menu

By clicking the **View menu**, the user can access the resource's metadata (raster and vector) or the alphanumeric data associated with the dataset (only vector). When "view data" is selected, the advanced search icons in the upper left corner of the table (Figure 9) provide the following options:

- perform an advanced search including attribute filtering, area of interest selection and layer filtering;
- filter the data displayed in the table according to the current view, after zooming to the desired area;
- zoom to page extent;
- hide or show columns in the table;
- synchronize the map with the filters applied to the data table.

Examples of queries using the "advances search" button are shown below.

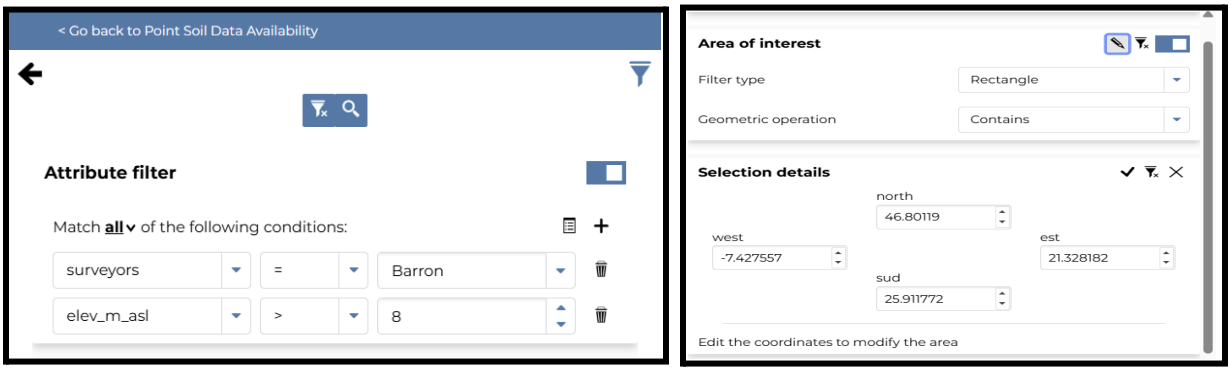


Figure 10 filters in the view data tool

By clicking the **Edit menu** (Figure 10), the user can edit dataset properties and perform the following operations:

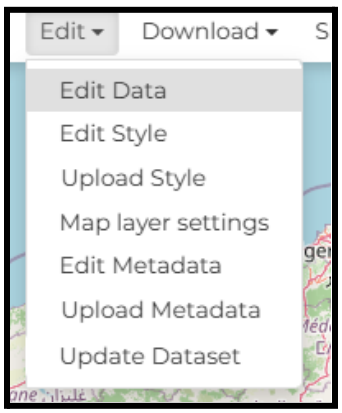


Figure 11: Edit Menu

- ***Edit data***: opens the tool for editing the data table, if available and if the user has "Edit" or "Manage" permissions.
- ***Edit St le***: opens the tool to changing the dataset style, if the user has "Edit" or "Manage" permissions. Knowledge of Style Layer Descriptor (SLD) standard is recommended.
- ***Load st le*** opens the tool for uploading an SLD style, provided that the user has "Edit" or "Manage" permissions. Familiarity with the Style Layer Descriptor (SLD) standard is recommended.
- ***Map la er setting*** allows the user to configure layer display options for maps
- ***Edit Metadata*** allows the user to edit the dataset metadata This information is particularly important for filtering datasets
- ***Upload Metadata*** allows the user to upload an XML file containing the dataset metadata. This operation should be performed with caution
- ***Update Dataset*** allows the user to replace the dataset by uploading a new file. This operation should be performed with caution.

By clicking the **Download menu**, the user can:

- download the dataset as a compressed **.zip** file containing the corresponding shapefile, which can be opened using GIS software;
- export the dataset in different formats including GeoJSON, GML-FeatureCollection, GML-GFS 1.1.0 FeatureCollection, Shapefile, and CSV;
- export only the portion of the dataset displayed in the current view using Crop dataset to current viewport;
- download only the subset defined by the applied filters using Download filtered dataset;
- download the ISO metadata in .xml format.

The “viewport” filter limit the search results d to resources that fall within the current map extent i.e. the visible area displayed on the screen.

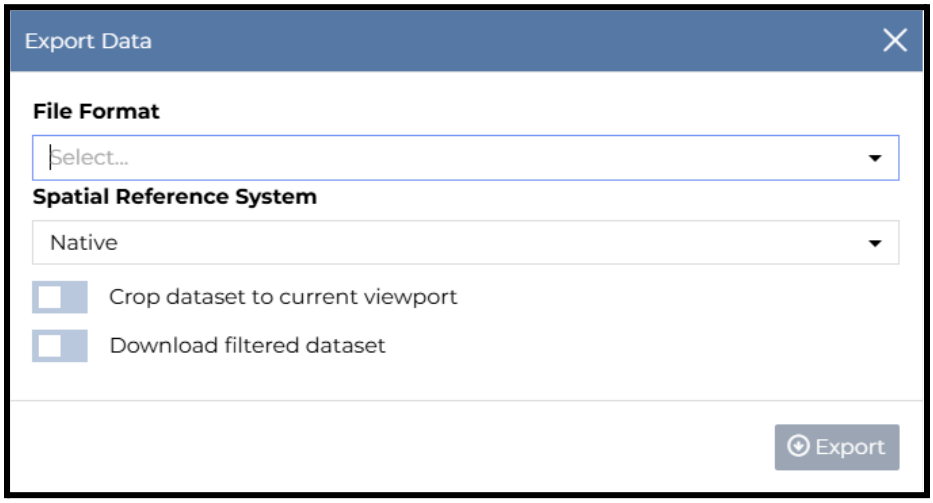


Figure 12: Download form

By clicking the **Share menu**, users with “Manage” s permissions can configure access permissions for the dataset. Using the permissions settings widget, they can define the following permissions for other users: "View", "Download", "Edit", or "Manage".

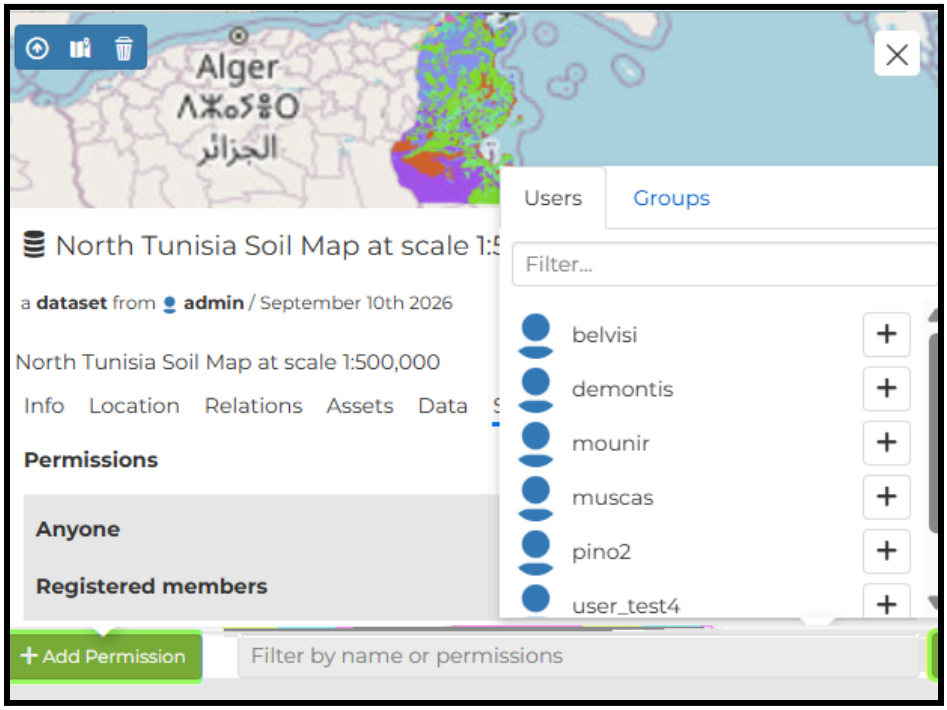


Figure 13: Permissions settings widget

To protect the dataset, it is recommended to set the permission level n to "none" for anonymous and registered users. Where appropriate, it is also more secure to assign permissions to individual users rather than to groups.

By clicking the **Filter menu**, the user can perform the same operations available through the icons in the **View menu.** The user can perform advanced searches including attribute-based filtering, and area of interest selection. For attribute filtering, multiple conditions can be applied by adding new conditions using the "+" button.

By clicking the **Create Map link**, the user can create a new **Map resource** using the selected dataset

By clicking the **print tool** the user can configure and generate a pdf

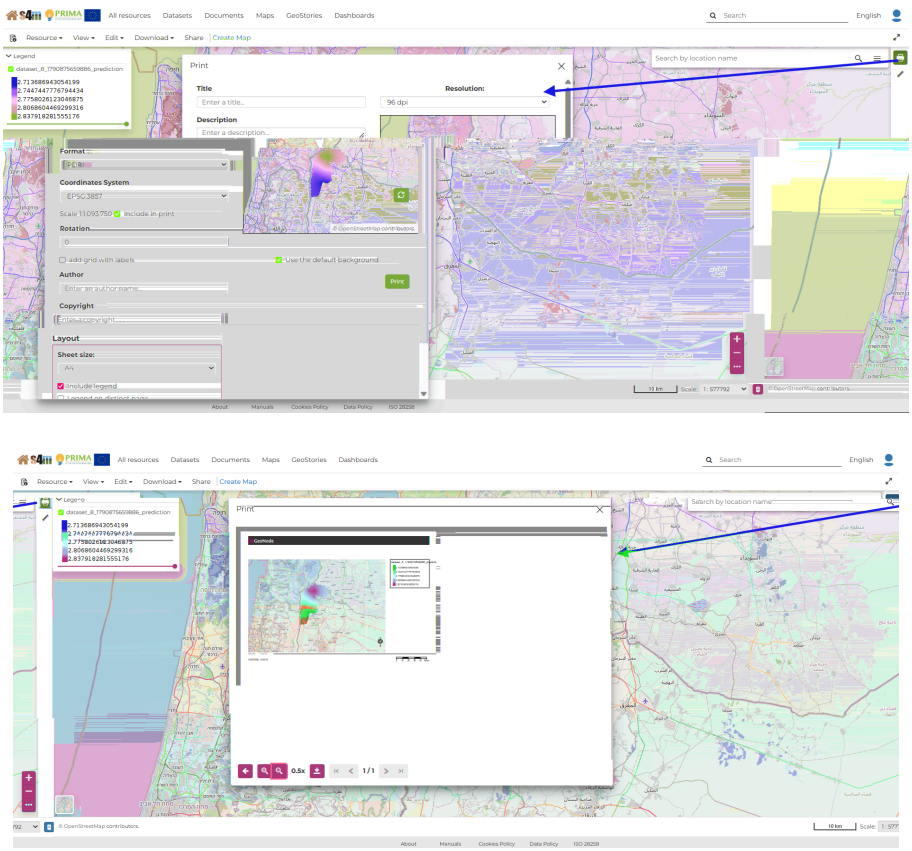


Figure 14: Print widget

Maps

Maps are compositions or visualizations built from ***one or more datasets*** (layers).

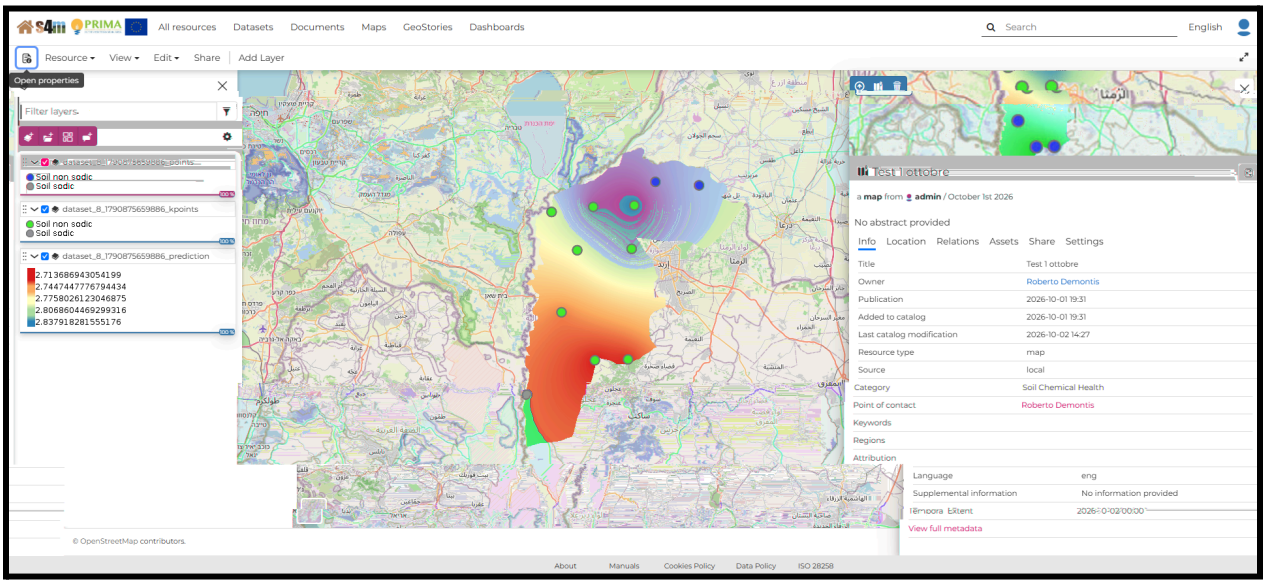


Figure 15, Map resource panel

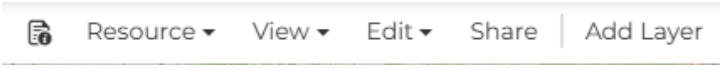


Figure 16 Map Resource menu

If the user has permissions to view a map and its associated datasets, they can perform the following actions:

by clicking the **Open Property button**, the user can view the resource's general information, geographic location, alphanumeric attributes, and linked resources that use or are used by it.

By clicking the **Resource menu**, users with staff permissions can save or delete the resource.

By clicking the **View menu**, the user can view the resource's metadata.

By clicking the **Edit menu**, the user can edit the map's metadata , which is particularly important for filtering resources or adding viewers.

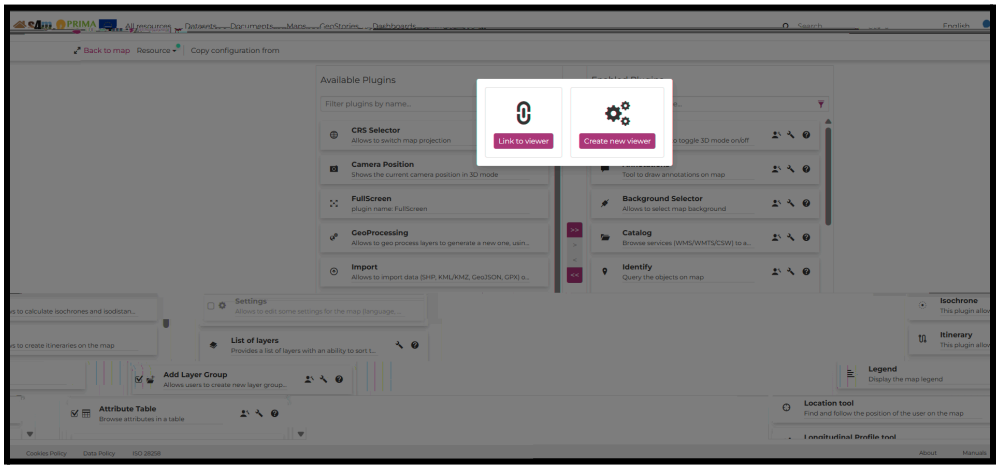


Figure 17, add viewer panel

By clicking the **Share menu**, users with “Manage” permissions can configure access permissions for the dataset. Using the permissions settings widget, they can define the following permissions for other users: "View", "Download", "Edit", or "Manage".

By clicking the **Add Layer link**, the user can add datasets to the map. Datasets can be selected from the catalogue or added from external datasets. External datasets can be configured by clicking the "+" button in the interface.

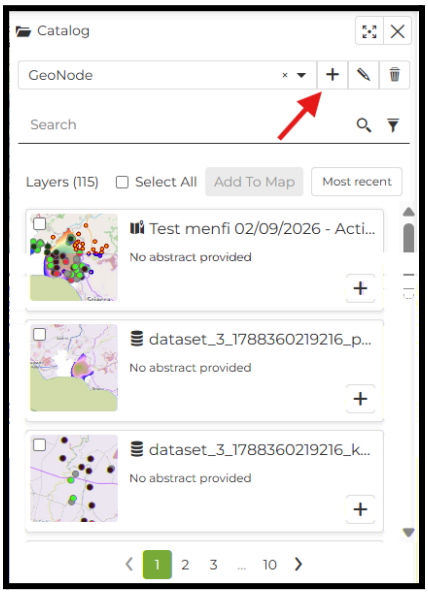


Figure 18, add external layer panel

The panel for exploring and configuring the map is shown in Figure 19

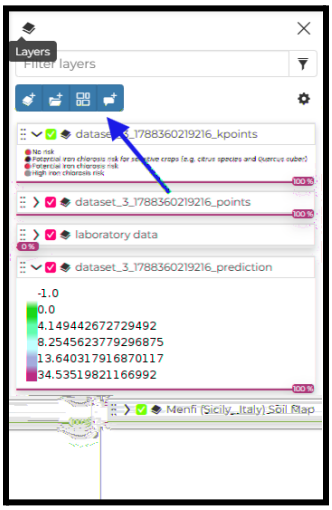


Figure 19, Map properties panel

By clicking the **first** of the four buttons on the left, data managers can add an external layer by selecting one of the following service types: WMS (Web Map Service), WMTS (Web Map Tile Service), CSW (Catalog Service for the Web), TMS (Tile Map Service), XYZ Tiles, WFS (Web Feature Service), and ArcGIS REST

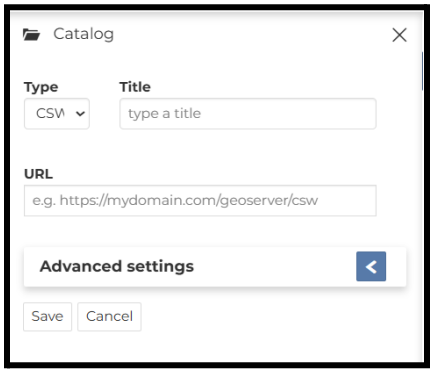


Figure 20, Add layer panel

The Advanced settings control how a remote layer is loaded, displayed, optimized, and interacted with:

- **Loaded**:
  - service search:automatically retrieves the available layers from the selected remote service
  - server type: defines the type of remote server, such as GeoServer or ArcGIS, to ensure that the appropriate requests are sent.
- **Displayed**:
  - preview: (displays a preview of the layer before it is added it
  - format: specifies the image format, such as PNG or JPEG, affecting image quality and

transparency
    - single tile vs tiled: determines whether the map is requested as a single large image rather than as multiple tiles. Single tile mode can provide better for quality but may result in slower performance
    - tile size: specifies the tile size, such as 256×256 pixels, affecting performance and the number of requests
- **Optimized**:
    - visibility limits: controls the zoom levels at which the layer is visible
    - tile grids: uses the tiling scheme defined by the remote service to ensure better alignment
    - domain aliases: specifies alternative domains for the same service, which can be useful for load balancing or browser related limitations
- **Interacted with**:
    - interactive legend: allows users to interact with the legend, for example by toggling styles or classes
    - information sheet: defines the format of the feature information displayed when a user clicks on the map, such as HTML, JSON.

By clicking the **second** of the four buttons on the left, users can add a new group of features to the current map.

By clicking the **third** and **fourth** buttons on the left, users can add annotations and widgets to the current map.

Users can typically add the following types of annotations: Points (markers), Lines, Polygons, Text labels / notes.

Users can customize annotation settings, including color, style (such as line thickness, fill, and icon), label and description.

If users have edit permissions, the annotations become part of the map. For generic users, annotations are usually temporary and are lost when the page is refreshed.

Documents

Documents are used to attach and share non-spatial files uploaded to the platform, such as: PDFs (reports, studies), Word documents, Excel spreadsheets, images, and presentations.

Users can open documents, for example by using PDF preview, and download them, depending on their permissions.

As with other SIS resources, documents may be opened, downloaded, edited, and managed according to the user's permissions. However generic SIS users can usually only:

- **View** the document and its metadata
- **Download** the document and the its ISO metadata

Geostories

GeoStories are a special content type used to create interactive narratives that combine maps and multimedia content. They are essentially web-based storytelling pages integrated: maps already available in the SIS, text such as descriptions and explanations), images and other media, and embedded content.

If the user has permissions to view the resources included in a geostory, they can:

- **view** the geostory and its metadata
- **scroll** through its sections/slides
- **read** text and view media content
- **interact** with embedded maps
- **explore** geographic content
- **follow** the narrative flow by navigating between sections, slides, or blocks.

Dashboards

Dashboards are interactive pages used to visualize and analyze geospatial data through charts, widgets, and maps. They are data visualization tools that combine: maps, charts (bar, pie, line, etc.), indicators/counters, filters, and widgets. Their purpose is to summarize and explore data in a dynamic and visual way.

If the user has permissions to view the resources included in a dashboard, they can:

- **View** the dashboard and its metadata, including information about charts, maps, and widgets
- **Interact** with visualizations
- **Apply filters** and view the corresponding dynamic changes
- **Explore** linked maps

If the user has permissions to edit and manage permissions, they can also add and configure widgets

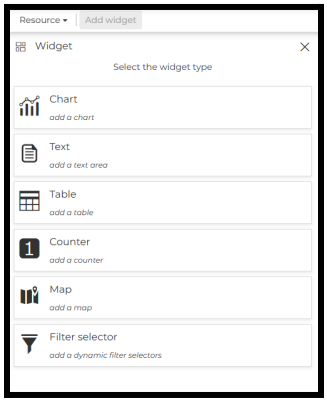


Figure 21, Add widget panel

3. Back Office

In addition to the standard catalogue functionalities for managing **soil data**, MED-SIS includes an additional tool known as the "**Back Office**."

Soil data are managed through the MED-SIS Back-office and published in the MED-SIS Catalogue

The Back Office is accessible only to data managers and SIS staff

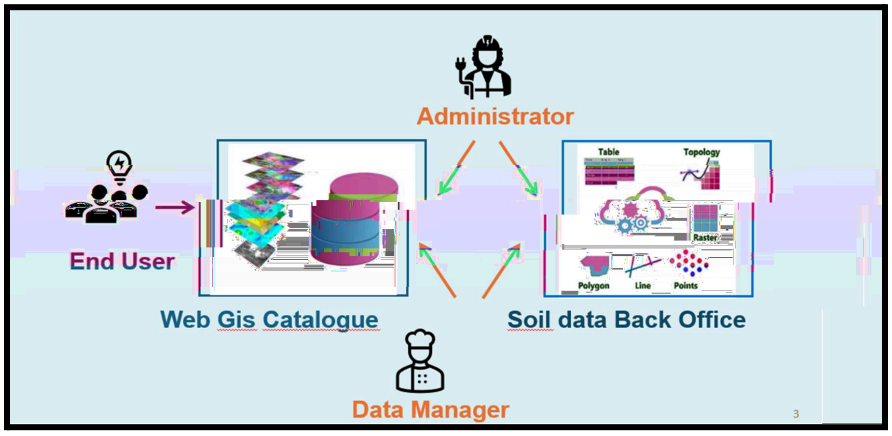


Figure 22, MED-SIS schema

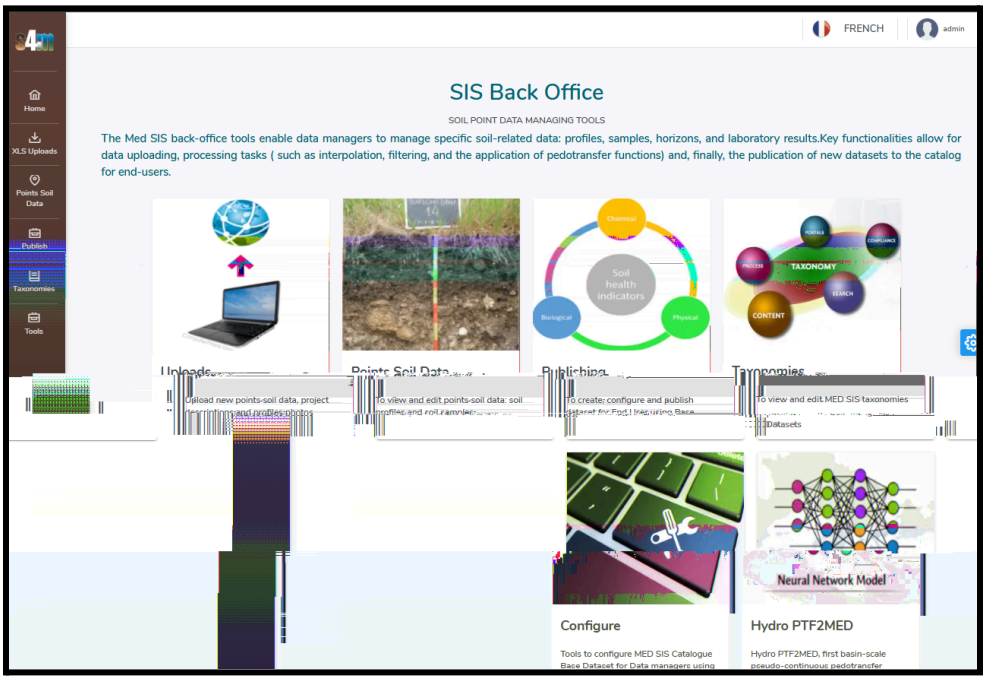


Figure 23, Back Office home page

Data Managers use Back Office tools to manage, filter, and process ***base datasets*** in order to create and publish **customized *datasets for end users***.

The Back Office homepage (Figure 23) displays the available tools. Each tool can be accessed by clicking on the corresponding description card:

- **Configure**: this is the first tool to be used after installation. It configures the **base datasets** and prepares them for publication in the catalogue
- **Upload**: allows users to upload new Soil Data and Metadata from XLSx files including point soil data, project metadata, profile photo metadata, and Laboratory Extra Measure data
- **Point Soil Data**: allows users to view and edit data relating to soil profiles and soil samples
- **Taxonomies**: provides information on all the taxonomies used for point-based soil data
- **Publishing**: enables the preparation, processing, and validation of datasets for end users
- **Hydro PTF2MED**: calculates a basin-scale pseudo-continuous pedotransfer function specifically developed for the Mediterranean region.

Configure

Following installation, MED SIS requires the configuration and publication of the "**base datasets**", which enable the automatic updating of soil data whenever new data is uploaded.

*The "**base datasets**" should be accessible only to data managers and Med-SIS Staff*

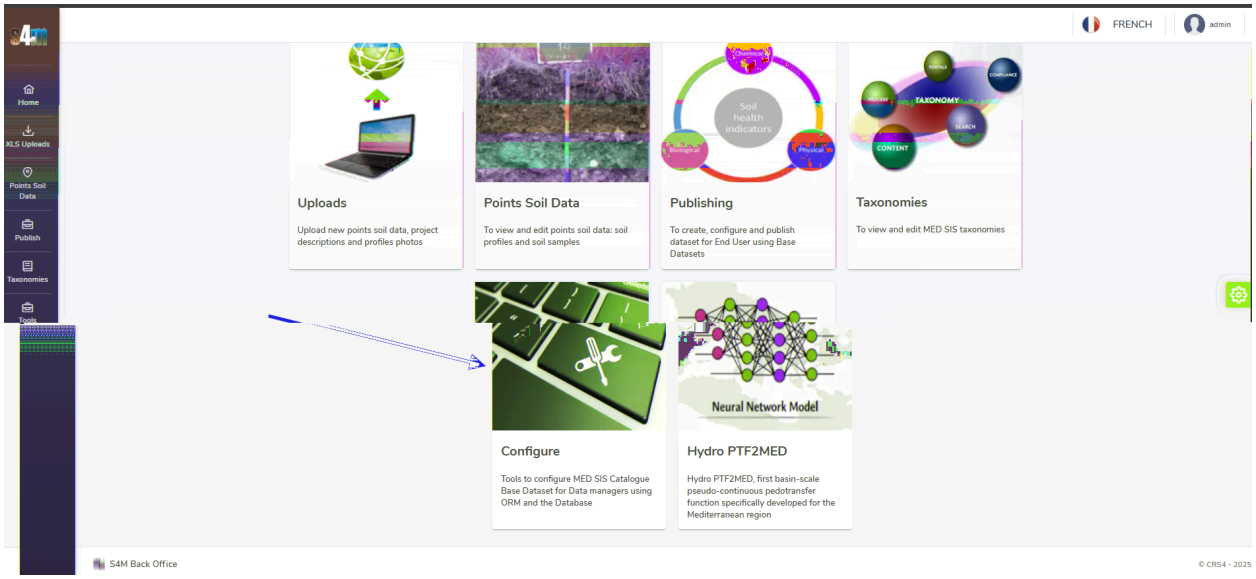


Figure 24: Back Office - Configure tool

The configure page shows two lists: **Soil Indicator base datasets** and Point **Soil Data Section Base Datasets.**

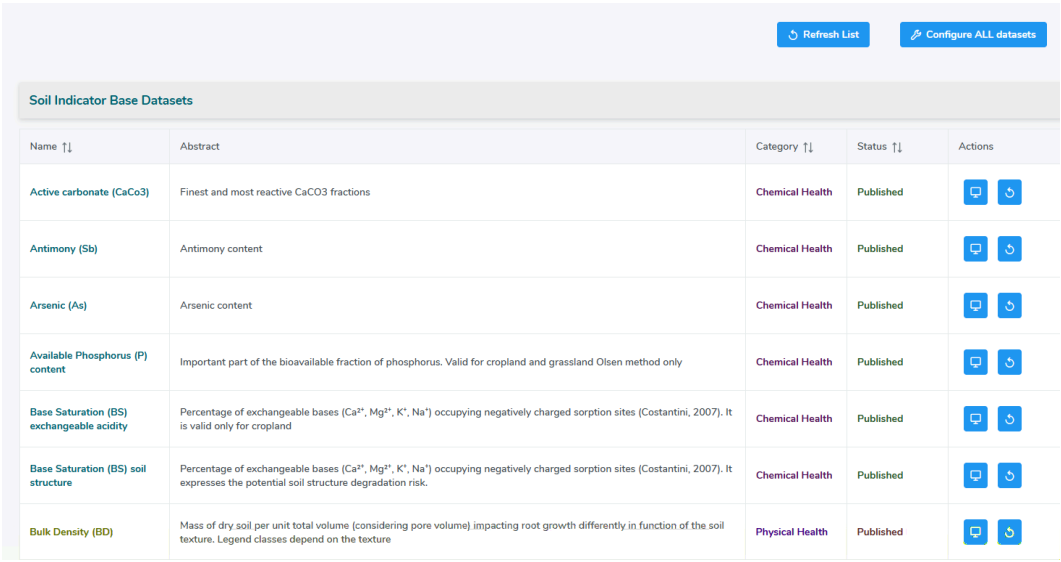


Figure 25: Soil Indicator base datasets

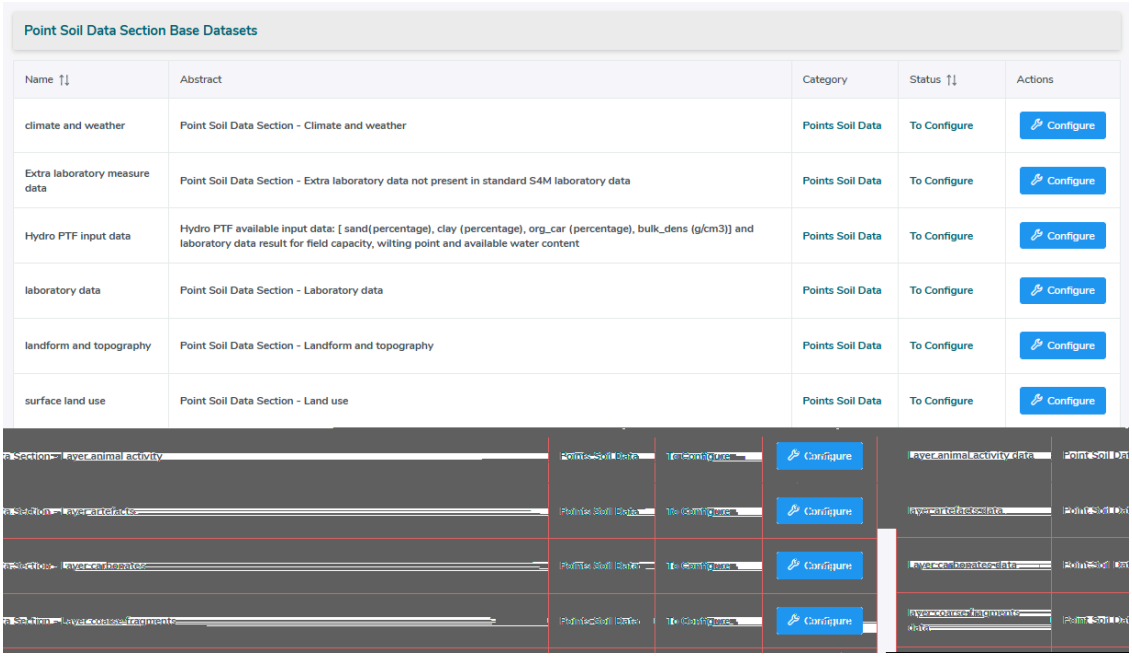


*Figure 26: Point Soil Data Section Base Datasets***.**

All the base datasets are by default loaded in the Geoserver but not in the Catalogue.

These datasets can be configured individually using the "Configure" button or collectively using "Configure all datasets" button.

Configuration **is required only once after installation**; it is performed on the server and may take several minutes and must therefore be carried out with caution.

Due to the number of datasets—approximately one hundred—using "Configure all datasets" may result in some errors. In this case, clicking the "Reconfigure" button can resolve the issue.

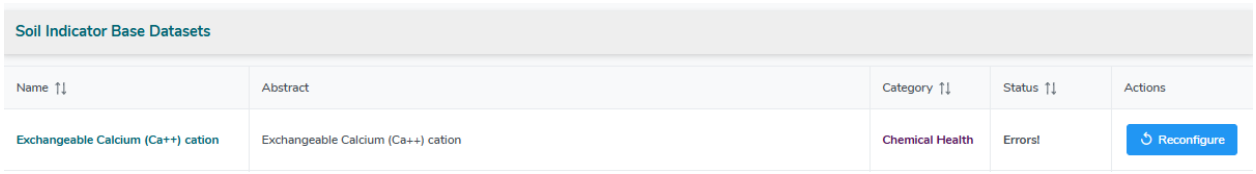


Figure 27: The reconfigure button

The published base datasets are shown in the Catalogue pages; It is recommended to edit their metadata

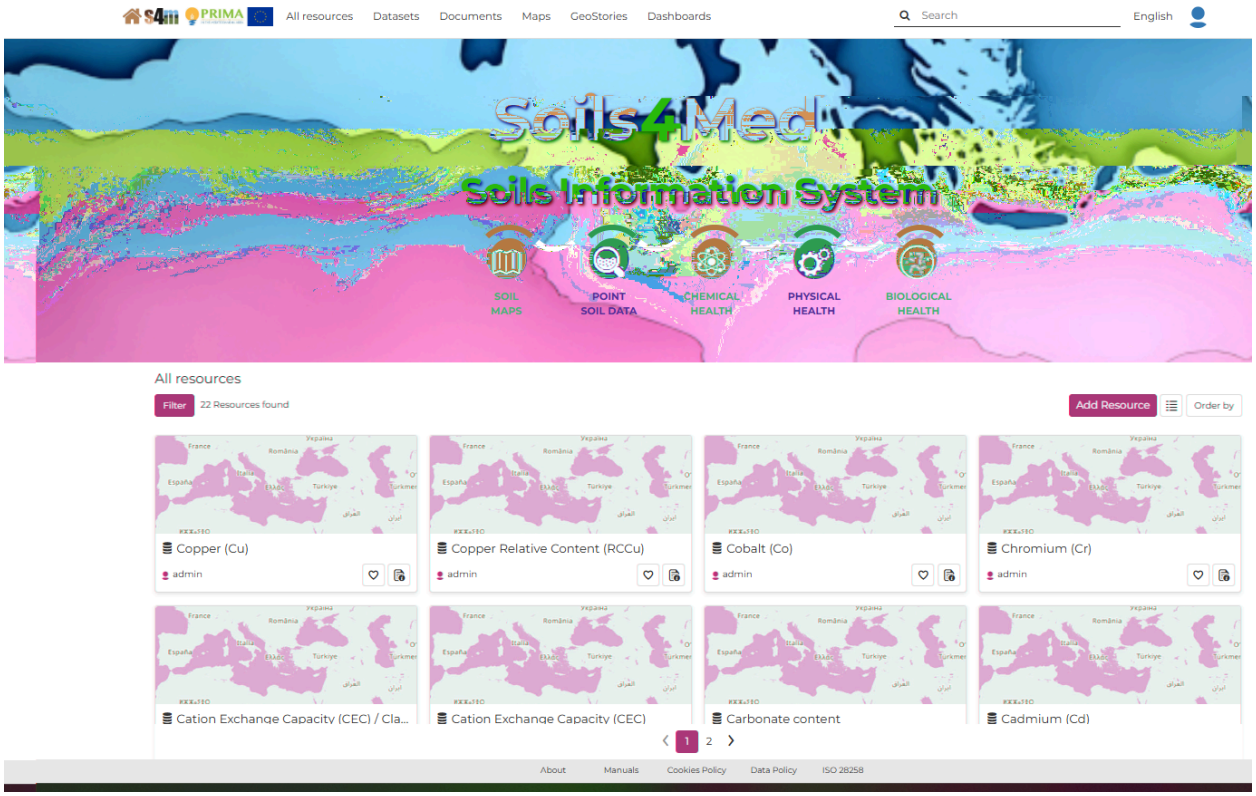


Figure 28: The list of base datasets

The datasets may be empty because they are automatically generated based on the uploaded data.

It is recommended to use the catalog pages for managing datasets, but certain operations can be performed in the administration panel, as shown in Figure 29.

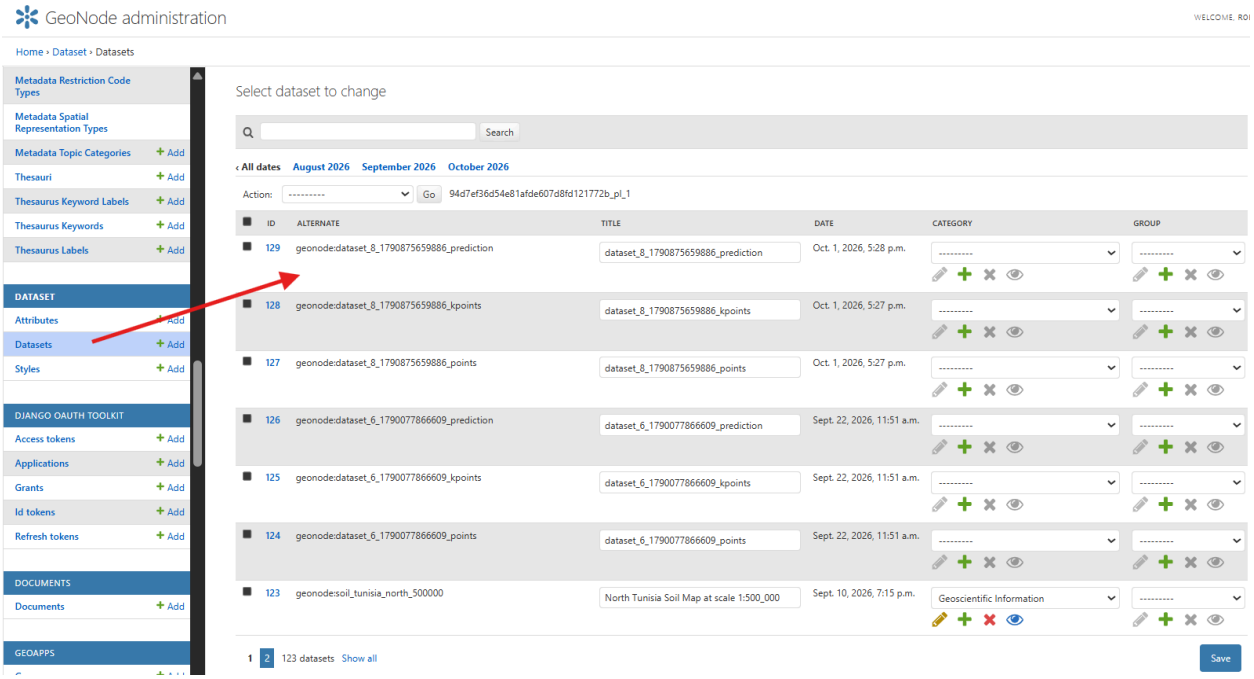


Figure 29: Admin Panel - Dataset Change

Upload

Data managers use the MED-SIS Back office to store soil data in the database. The primary soil data uploaded to MED-SIS are point-based soil data, which include information on soil profiles and samples. Other types of information required by MED-SIS include: project metadata, photo metadata and laboratory extra measures, i.e. measurements not included in the default data template.

The data must be uploaded in **XLSx format**. A template is available for each type of information in the doc folder of the MED-SIS GIthub repository (https://github.com/crs4/med_sis/tree/main/src/s4m_catalogue/static/doc).

Specifically, point-based soil data are managed as catalogue resources, form part of the "**base datasets**" and provide the input for the automatic generation of the soil indicators, which constitute another component of the MED-SIS data infrastructure.

For these activities, the process typically starts with an Excel file complaint with one of the templates available for download from the back office.

The *XLS Uploads* button on the back office home page opens the *List* of files that have already been uploaded to the SIS. The list displays the relevant details, including file name, the user who performed the upload, the upload date, the file type, the action performed, and the outcome (Status).
The following actions are available:

- By clicking the delete icon, the user can remove the upload entry from the list. This does not delete the data imported into the database.
- By clicking the information icon, the user can open the upload metadata panel
- By clicking "**New XLSx upload**", the user can open the "*New XLSx upload* " panel

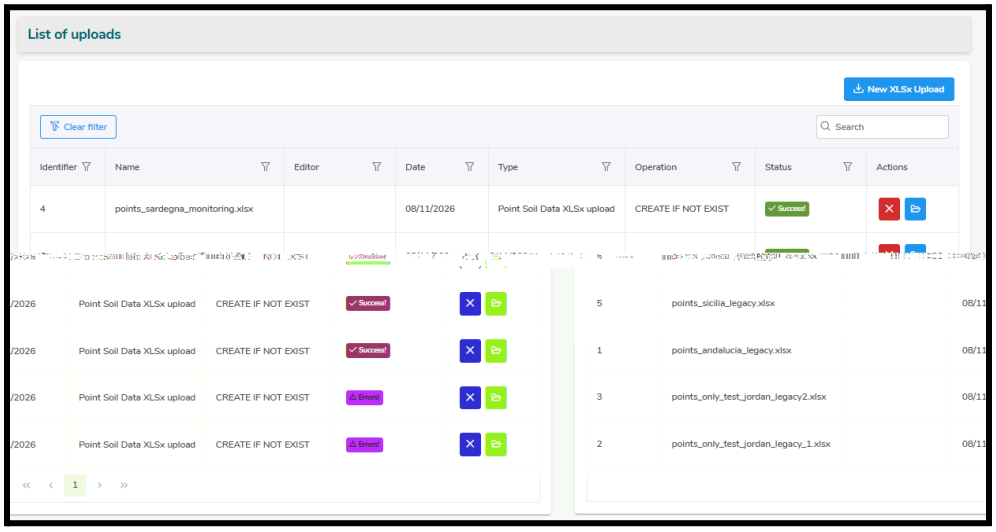


Figure 30 List of uploaded dataset

An Upload can have one of the following four states:

- **ELABORATING**: the upload is being processed
- **SUCCESS**: the dataset has been successfully uploaded to the server
- **ERRORS**: the dataset has been uploaded to the server, but errors were detected during process,
- **CRITICAL ERROR**: the upload has generated system errors

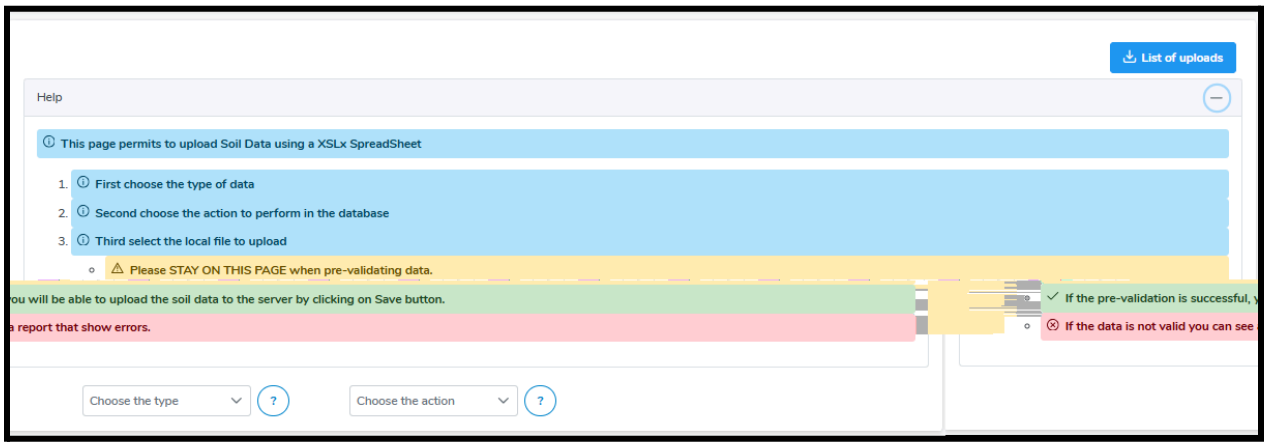


Figure 31 New XLSx upload panel

The *New XLSx upload panel* display several help sections, either directly in the panel by clicking the help icon

The data manager first selects the data type and the action to be performed. The available data type are: Point Soil Data, Laboratory Extra Measure, Project Metadata, and Photo Metadata.

The user then selects the required database action:

- ***CREATE IF NOT EXIST*** for each item in the uploaded file, MED-SIS attempts to insert it into the database. If an item with the same id already exists, an error is generated.
- ***REPLACE IF EXIST***: for each uploaded item, MED-SIS attempts to replace the corresponding item in the database. If the item does not already exist, an error is generated.
- ***UPDATE IF EXIST***: for each uploaded item, MED-SIS applies partial changes to the corresponding fields in the database. If an item with the same id does not exist, an error is generated.
- ***DELETE IF EXIST***: for each uploaded item, SIS attempts to delete it into the database.

For each data type, users can click the question mark icon next to the options list to download the official .XLS template and the instructions for filling it in. Once the options have been selected, the "Choose file" button becomes available.

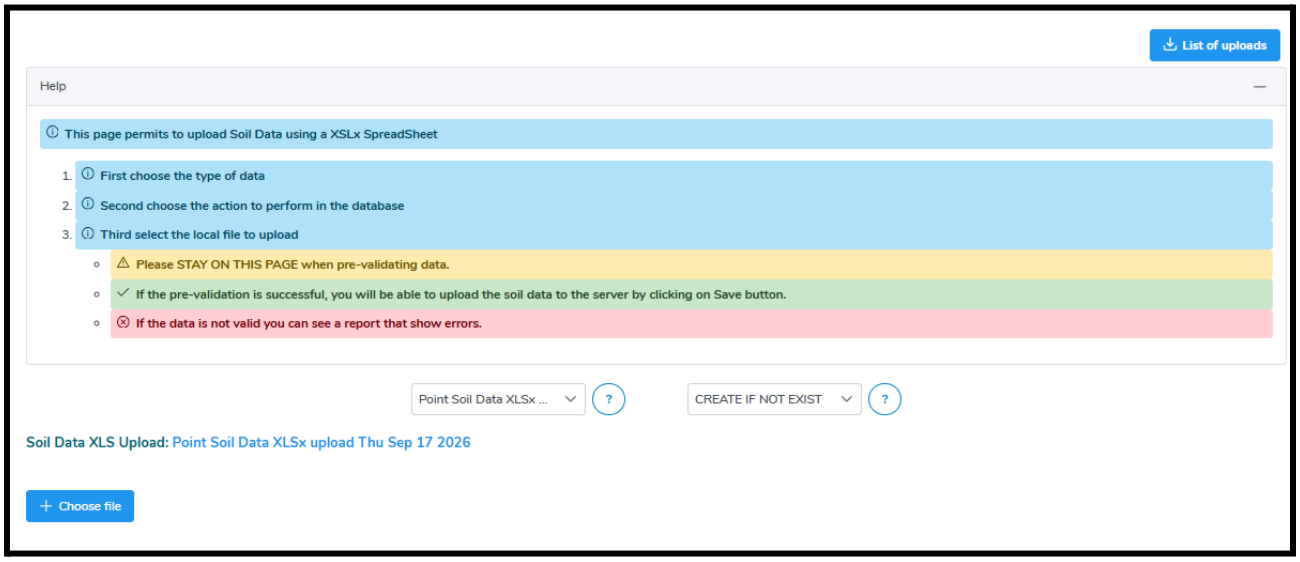


Figure 32, Upload tool page

After the XLS file has been successfully uploaded, a confirmation message indicates that the file has been successfully converted to a GeoJSON format. Users can scroll down to view the uploaded points on a map. If no errors occur during the upload, a No Errors message is displayed below the map, as shown in Figure 33.

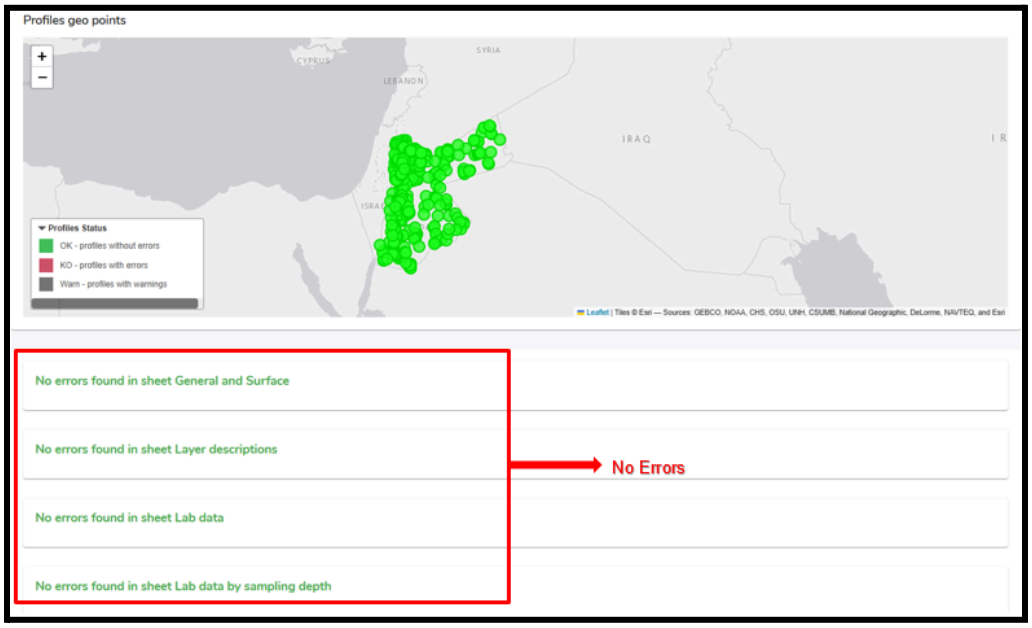


Figure 33 Dataset chosen and errors preview

Users can then: click ***Reset*** button to restart the upload process,click ***Save Data*** to upload the data and perform the action on the database. The Save Data operation may take several minutes to complete.



Figure 34 Dataset chosen and "Save Data" button

If errors occur during the upload, users can view a log reporting the type of error and the corresponding cell (row and column) in the XLSx file.



Figure 35 Dataset chosen with errors

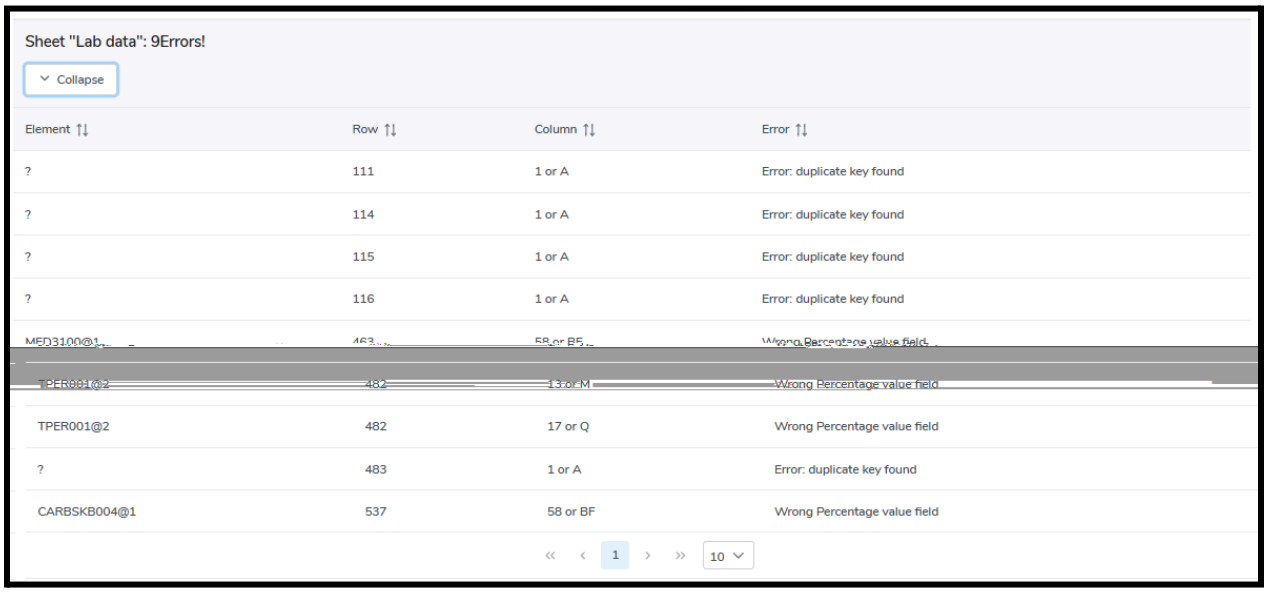


Figure 36 Errors log

The Point soil data Excel file

According to the *MED-SIS-TEMPLATE_POINT_SOIL_DATA.xlsx* template, the spreadsheet must contain four sheets:

General and Surface

La er description

Lab data

Soil Classification

The data in General and Surface and Layer description must start from rows 10 and 6, respectively. The data in Lab data and Soil Classification must start from row 3.

Before uploading the file, make sure that all data in the Excel file are stored as 'values' and that no fields contain formulas or text formatting rules. Otherwise, MED-SIS may return errors*:*

Error Message table

Error msg	*Description*
Wrong Classification code field	*The classification code entered in the cell is not recognized by the SIS.*
Wrong Number field	*The value entered in the cell is not a number* *Check whether the cell contains a formula instead of a value*
Wrong Boolean field	*The value entered in the cell is not a Boolean value.* *Allowed values: T for True, F for False'*
Wrong Date field	*The value entered in the cell is not a valid date. The allowed format is DD/MM/YYYY* *Check whether the cell contains a formula instead of a value*
Wrong Percentage value field	*The value entered in the cell is not a valid percentage.* *Check whether the cell contains a formula instead of a value*
Wrong Positive value field	*The value entered in the cell must be a positive number or 0.* *Check whether the cell contains a formula instead of a value*
Wrong Latitude value field	*The latitude must be expressed in decimal degrees.* *Check whether the cell contains a formula instead of a value*
Wrong Longitude value field	*The longitude must be expressed in decimal degrees.* *Check whether the cell contains a formula instead of a value*
Error found Rich Te t	*The cell contains text formatting rules. Please use plain text instead*
Error duplicate ke found	*The element identifier is the same as the identifier of another element in a previous row*

Point Soil Data

The Point Soil Data tool allows users to perform CRUD operations (Create, Read, Update, and Delete) on individual soil profiles or samples.
As shown in Figure 37 the first page of the tool displays a list of soil profiles and samples, together with a set of options that allow users to search for and identify specific elements in the list.

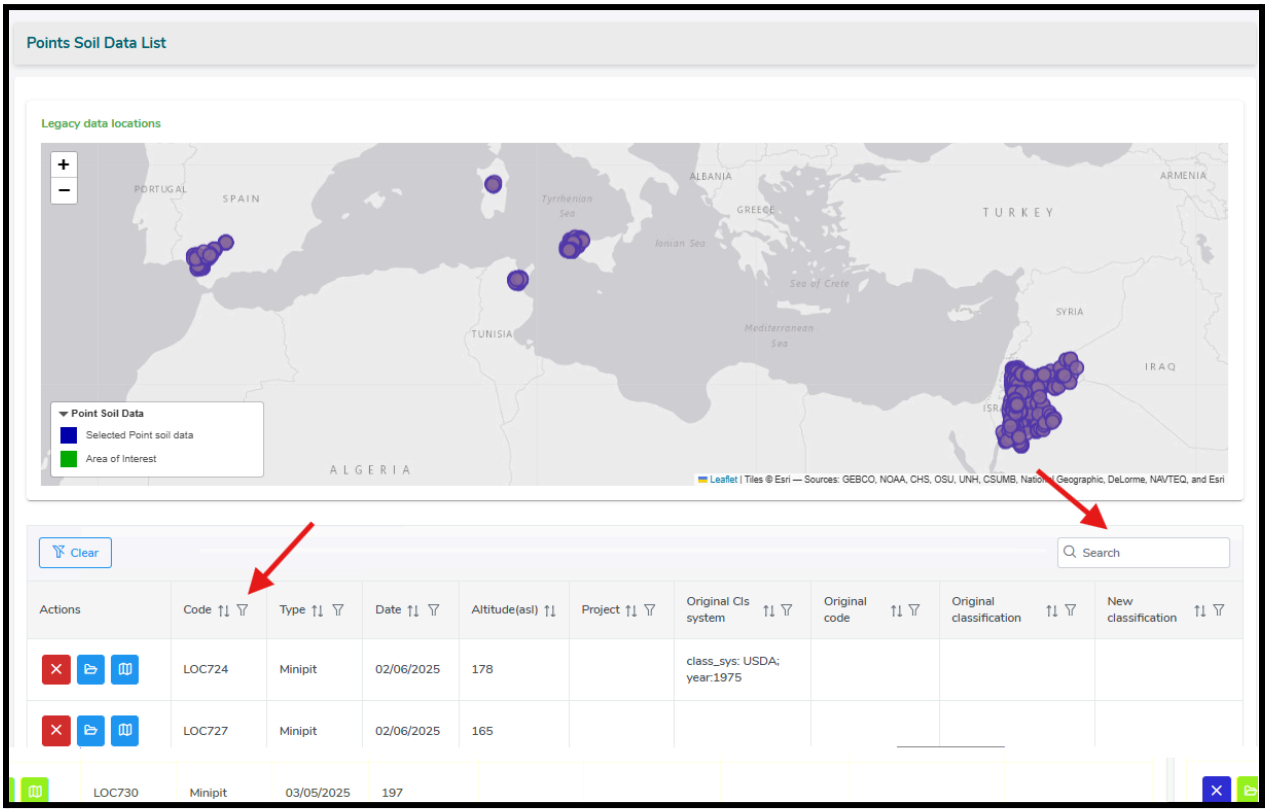


Figure 37 Point Soil Data list

The following actions are available:

By clicking the delete icon, the user can delete the selected record from the list. This operation also deletes the corresponding imported data.

By clicking the details icon, the user can open the **Point Soil Data panel** for the selected element.

By clicking the map icon, the user can display the selected element in the map panel.

The Point Soil Data panel (Figure 38) displays all the information associated with a soil profile or sample.

The red box indicates the summary card describing the profile / sample. The central panel displays the depth and Munsell colour of the profile layers, while the panel on the left shows the profile location.

The red arrow points to the sections containing the profile data. A Profile can potentially contain around one thousand fields.

The fields are organized into sections. Each field has a name, a data type and an editable value.

By clicking the edit icon, the user can edit the value of the field

By clicking the information icon, the user can open the information panel associated with the field

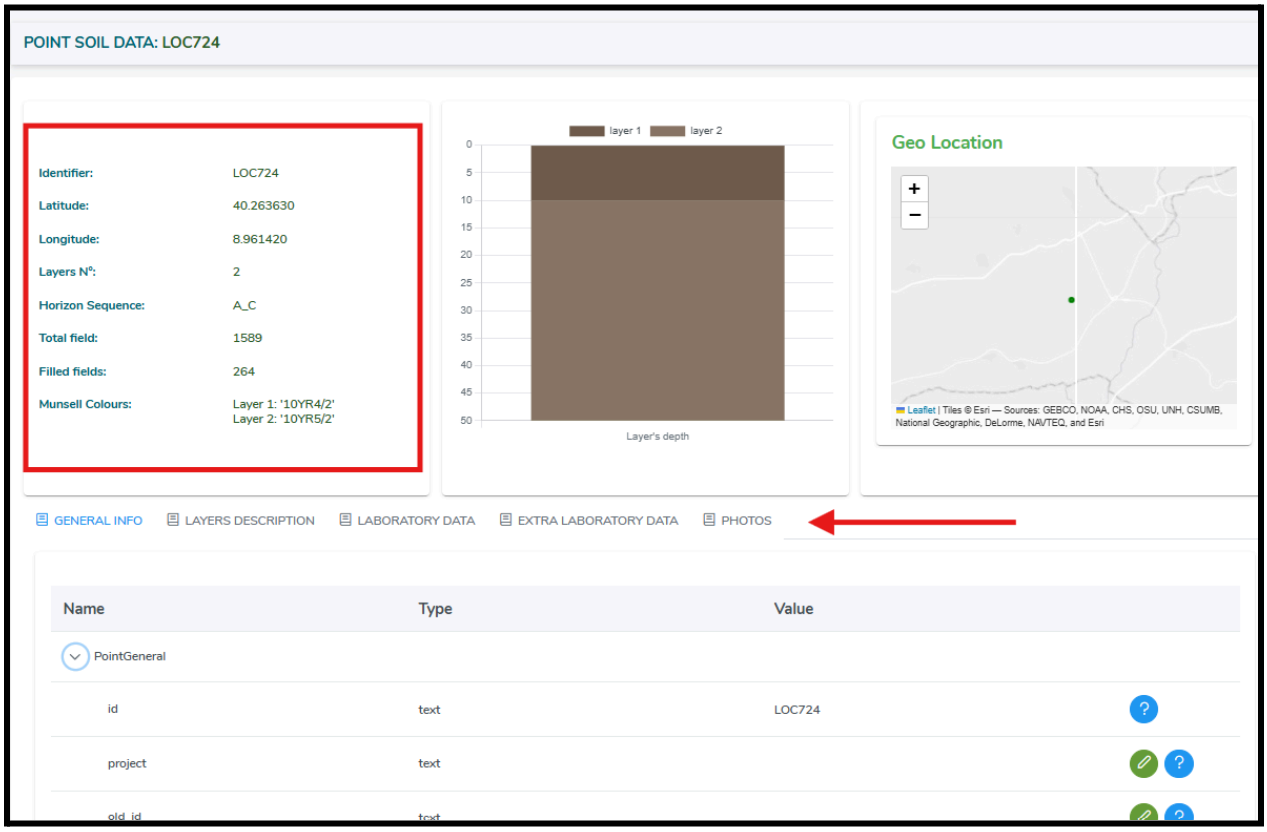


Figure 38 Point Soil Data view

Taxonomies

The Taxonomies tool allows users to explore all the taxonomies used in the creation and management of soil data.

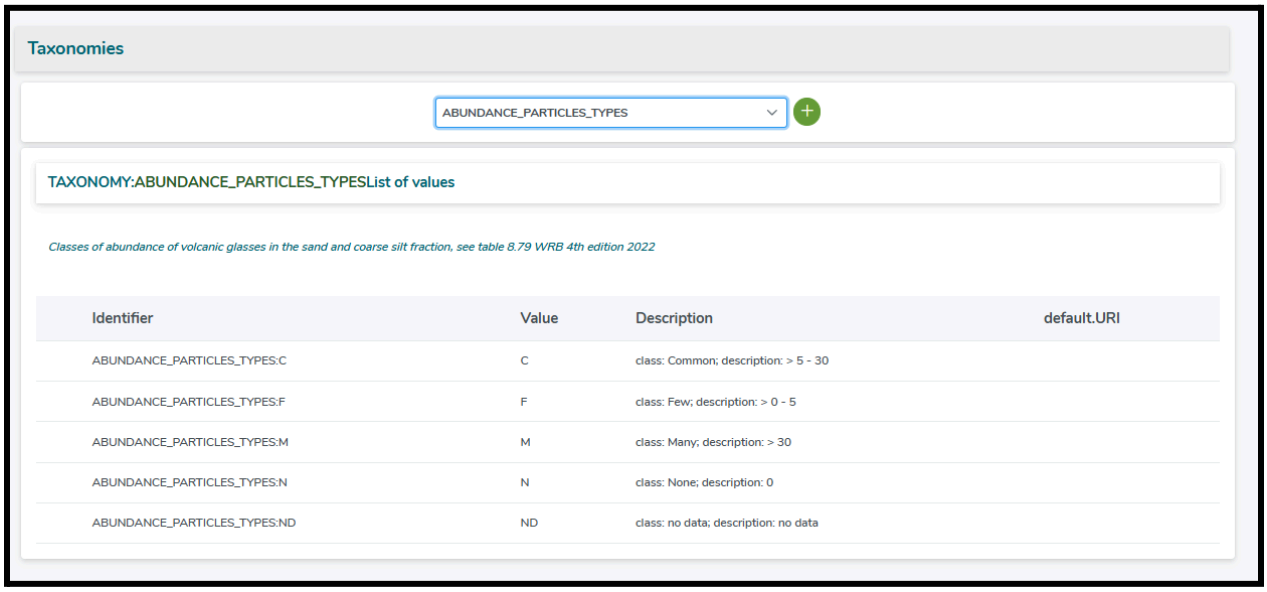


Figure 39 Taxonomy list

The drop-down list allows users to select a taxonomy, which is then displayed as a list.
Each item in the list describes a taxonomy entry using: the following fields:

- **identifier**: the identifier used in the database
- **Value**: the name or value of the entry
- **Description**: a description of the entry
- **URI**: an external URI referring to an international standard vocabulary, such as https://qudt.org/vocab/unit/MilliGM-PER-KiloGM) .

By clicking the icon, users can open the panel for creating a new taxonomy. This may be necessary when additional laboratory measurements need to be entered.
These operations must be coordinated with the other information systems (SISs) in the network to ensure **interoperability** and data sharing.

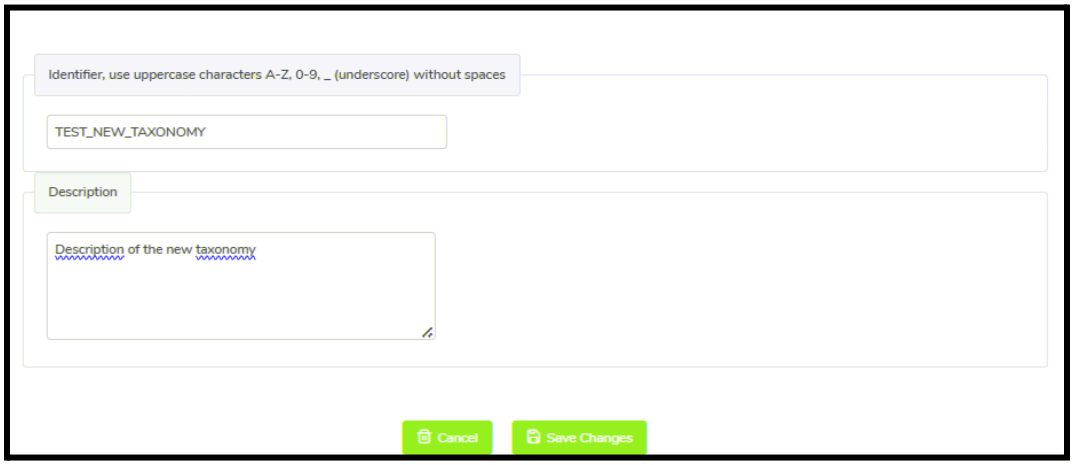


Figure 40 Taxonomy create

Once a taxonomy has been created, new entries can be added to it.

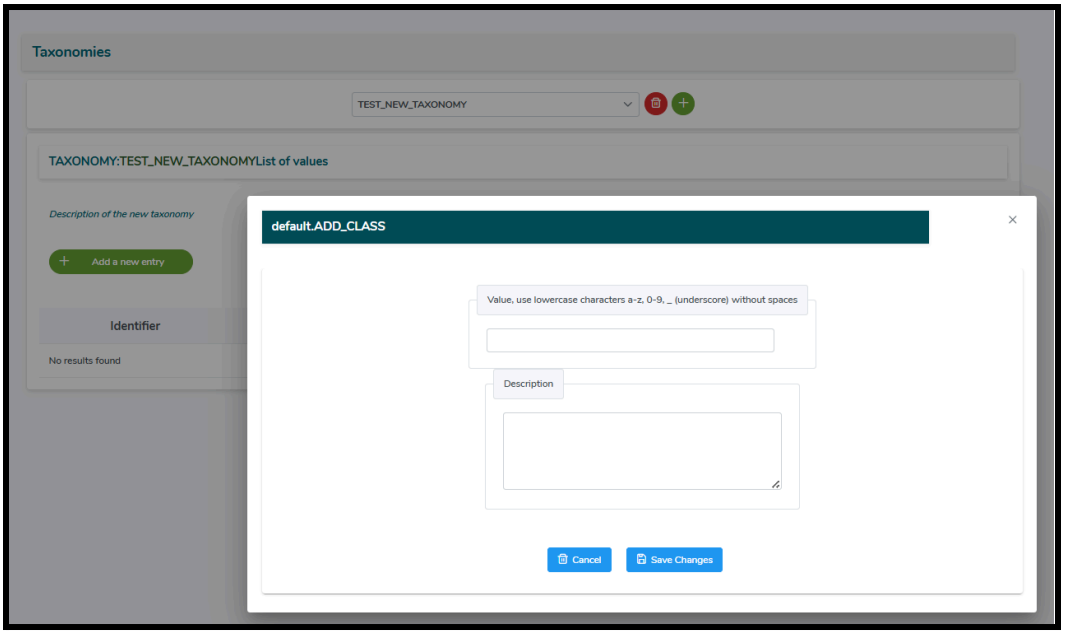


Figure 41 Taxonomy - add classification

Taxonomies can also be deleted by clicking the the **delete** icon.

To ensure interoperability, **default taxonomies cannot be removed or modified**.

Publishing

The *Publishing* tool, accessible from the back office home page, is the MED-SIS **main tool** for create **customized datasets for end user**

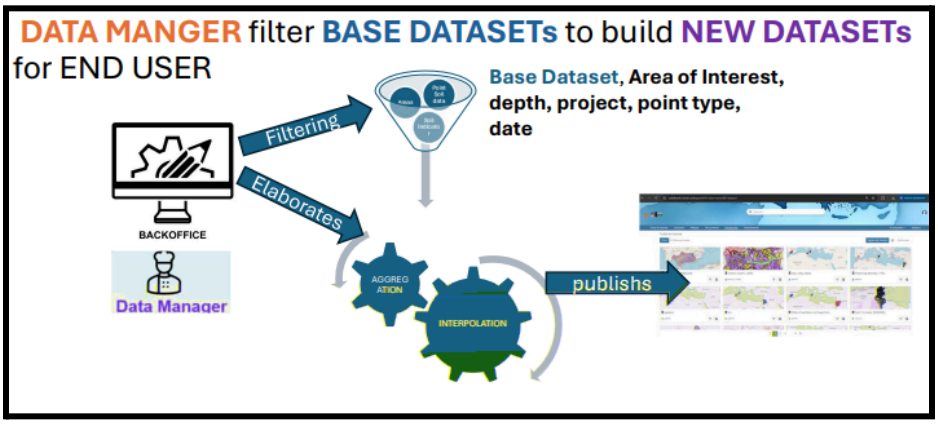


Figure 42 Publishing data flow

The first page displays the *list* of datasets that have already been created, together with relevant details and icons for the available actions.

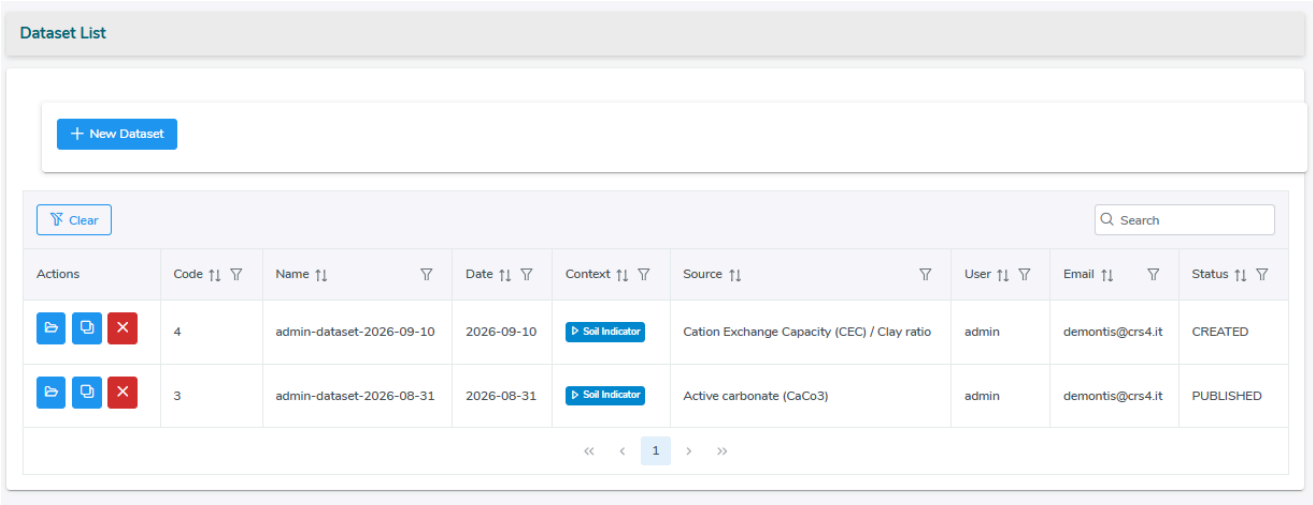


Figure 43 Publishing - customized dataset list

By clicking the delete icon, the user can remove the dataset entry from the list. **This does not delete the corresponding published datasets from the catalogue**.

By clicking the details icon, the user can open the dataset panel, which displays the dataset details and current status.

By clicking the "**New Dataset**", the user can open the "*New Dataset*" panel (Figure 44)

There are three types of customized dataset, depending on the type of source base dataset:

- Point Soil Data Sections: based on a source Base Dataset of type Point Soil Data
- Soil Indicators: based on a source Base Dataset of type Soil Indicator
- Soil indicators evaluated in an area: based on a source Base Dataset source of type Soil Indicator

When a dataset is created, it appears in the dataset list with the status "CREATED"

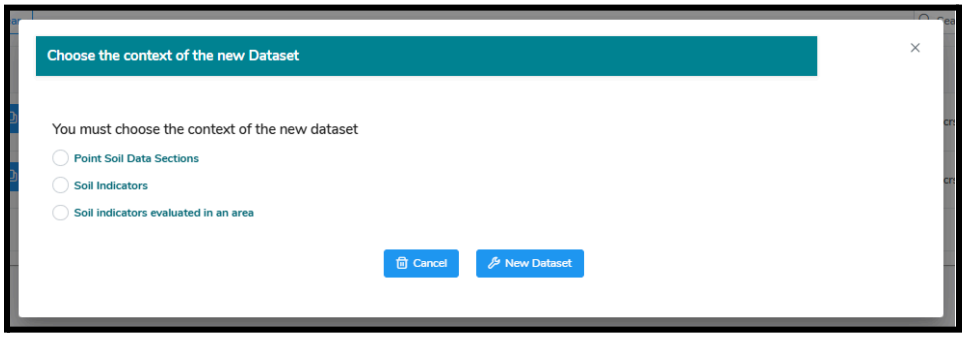


Figure 44 Publishing - customized dataset types

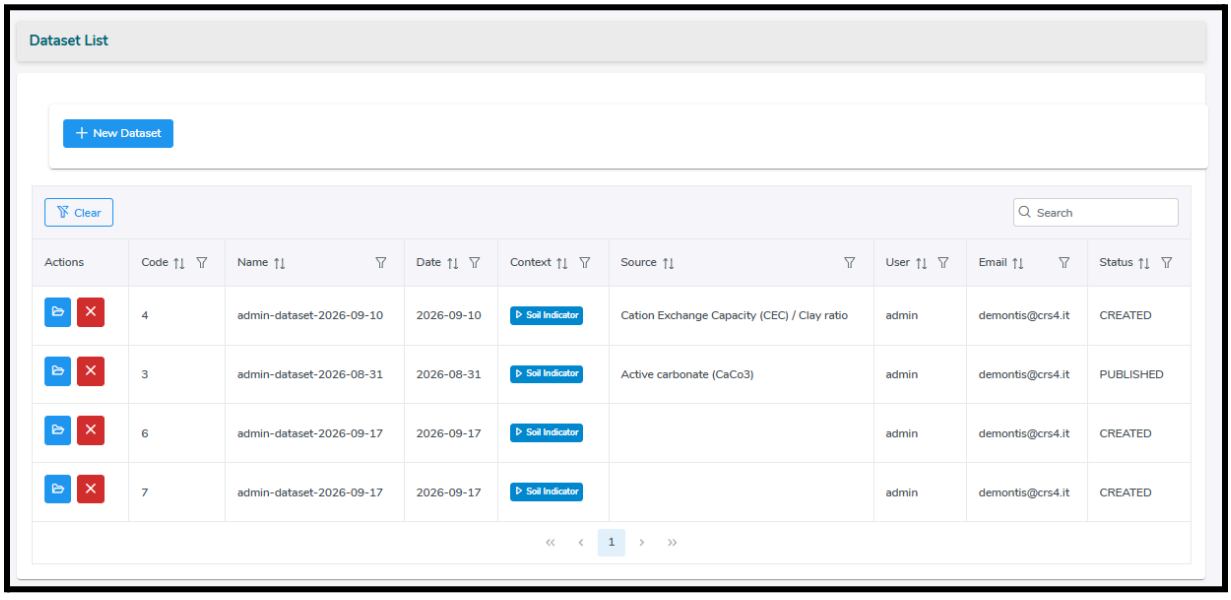


Figure 45 Publishing - customized dataset types

A dataset can have four possible states:

- **CREATED** : the dataset has been created but not configured

- **CONFIGURED** : the dataset has been configured but not published in the catalogue
- **IN_PROCESS** : the dataset is being processed
- **VALIDATED** : the dataset has been validated by the data manger,
- **PUBLISHED** : the dataset has been published in the catalogue,
- **ERRORS** : errors occurred during dataset processing.

The data manager uses the tool to progress the dataset the following workflow: :

CREATED => CONFIGURED => IN_PROCESS => VALIDATED => PUBLISHED

By clicking the details icon the user opens the dataset panel, which displays the current status of the dataset.

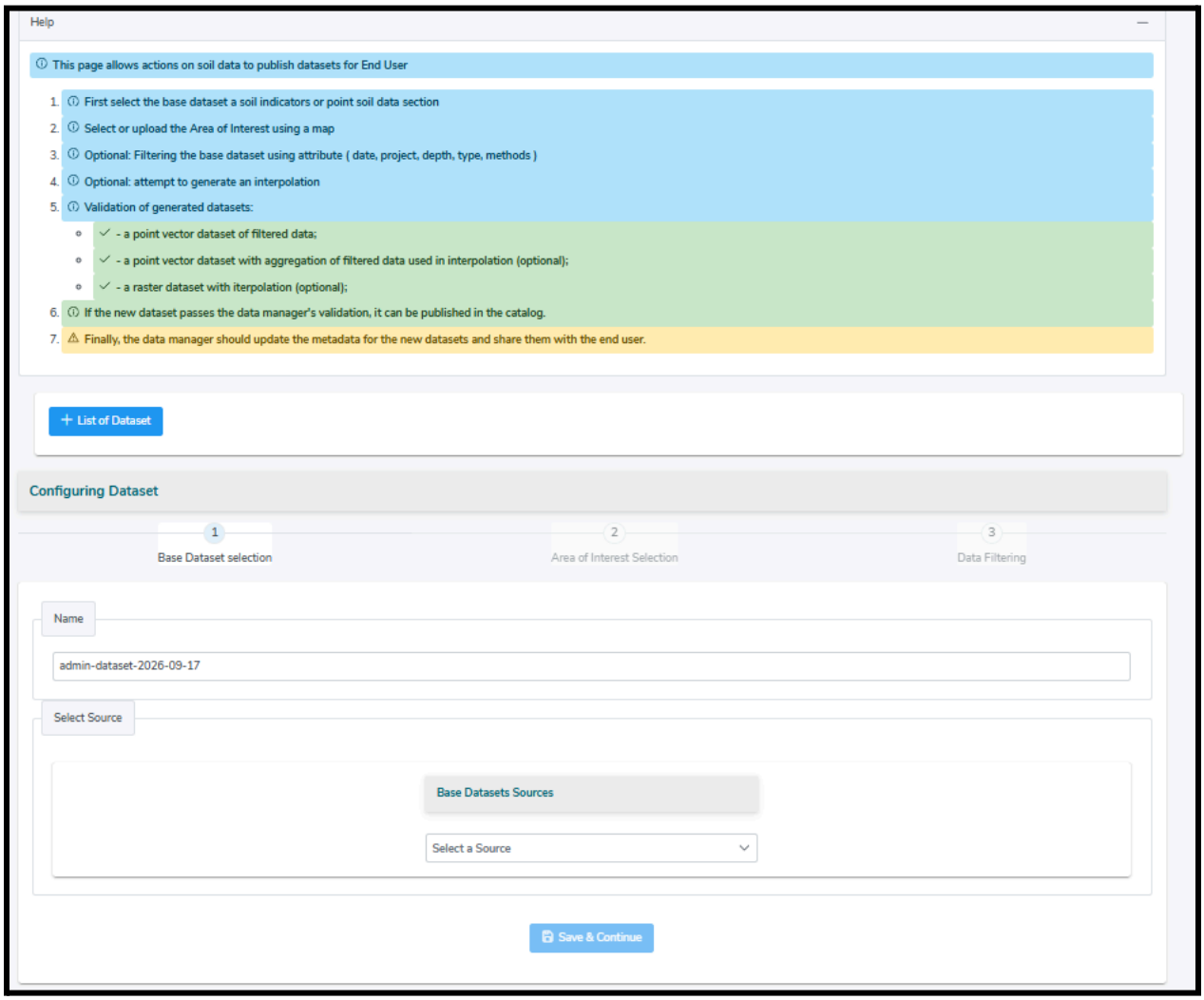


Figure 46 Publishing - dataset configuration

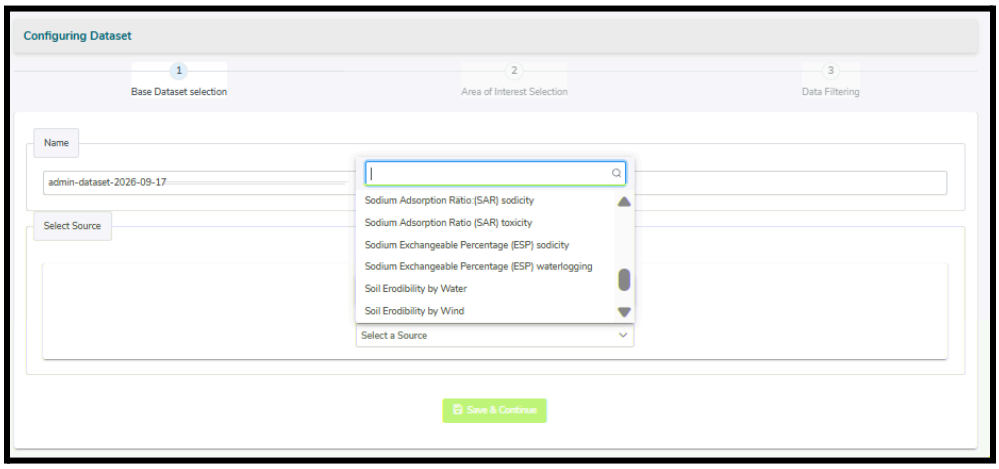


Figure 47 Publishing - Configuration Step 1. Base Dataset Selection

After selecting the source **Base Dataset**, the system loads **all the points** contained in the selected Base Dataset. The user can then click **Save & Continue**.

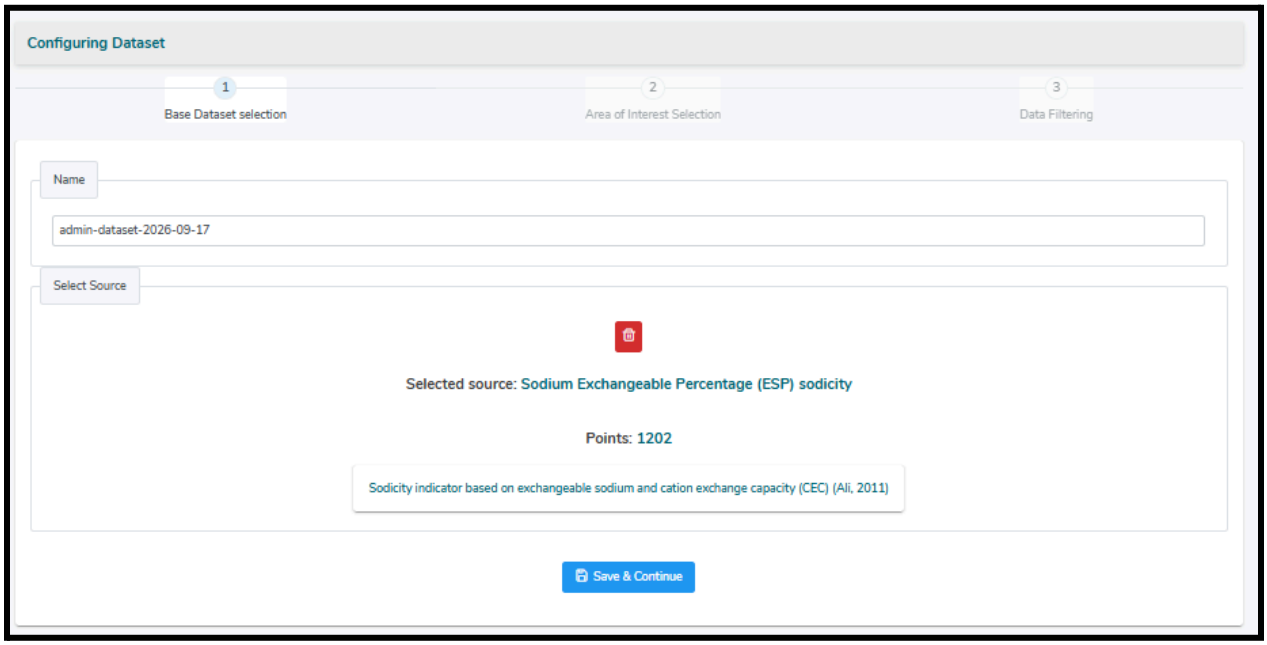


Figure 48 Publishing - Configuration Step 1. Base Dataset Selected report

The next step is to select the Area of Interest (AOI). The AOI can be defined either by drawing a custom polygon (Figure 49) or by selecting polygons from a Boundaries Catalogue dataset (Figure 50)

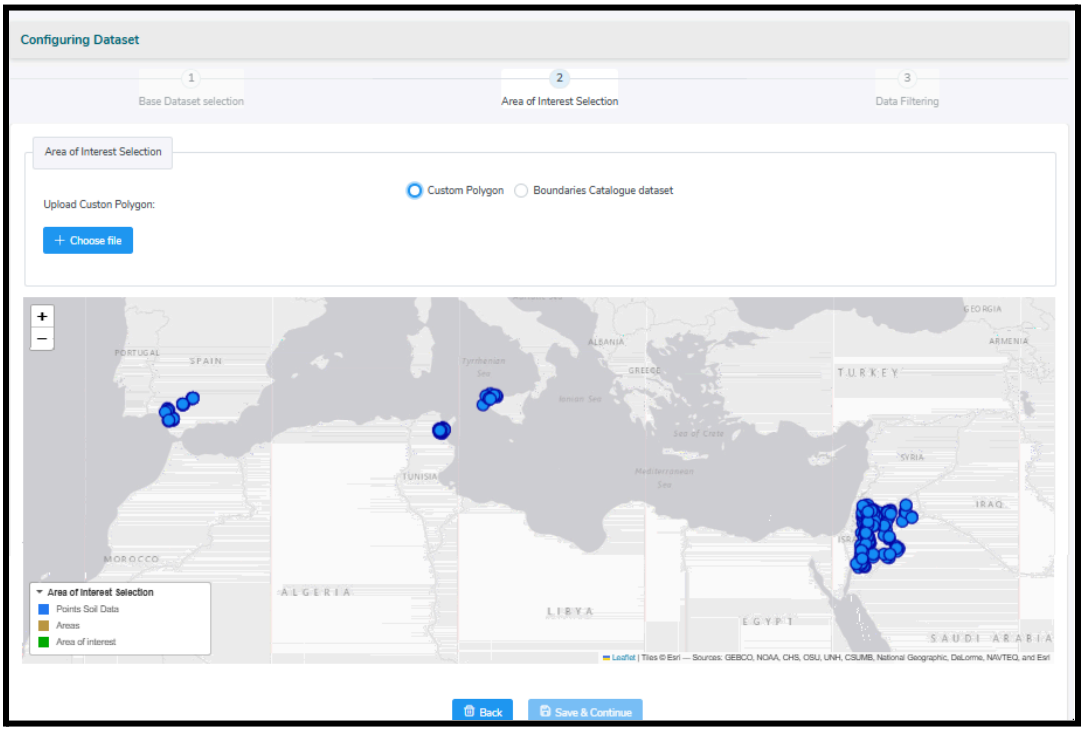


Figure 49 Publishing - Configuration Step 2. Custom polygon as AOI

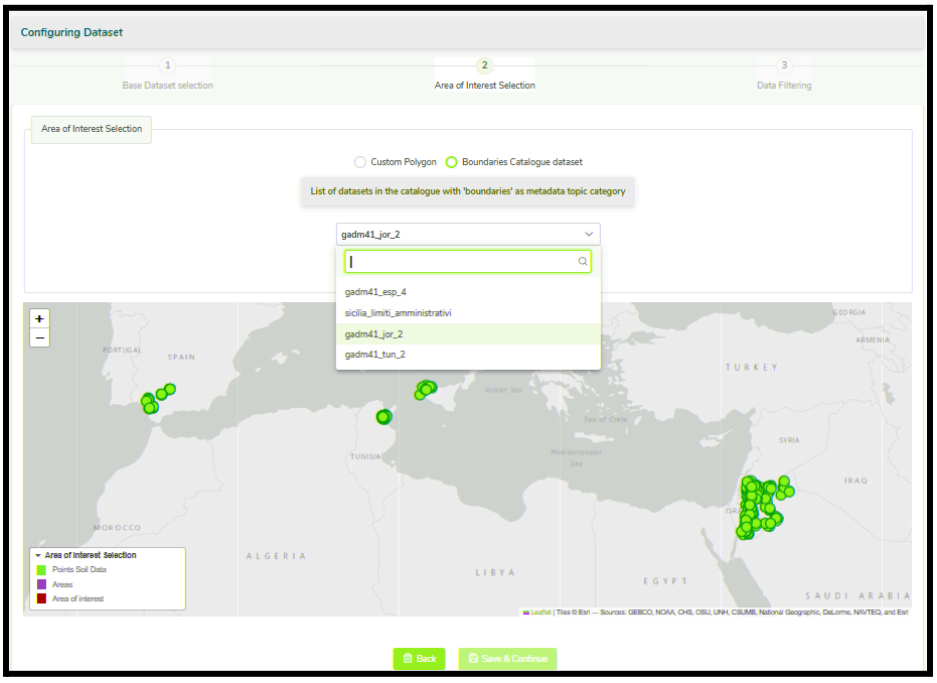


Figure 50 Publishing - Configuration Step 2. Catalogue resources as source for AOI

To be used as a "Boundaries" dataset, a catalogue dataset must have its "Topic category" metadata field set to "Boundaries".

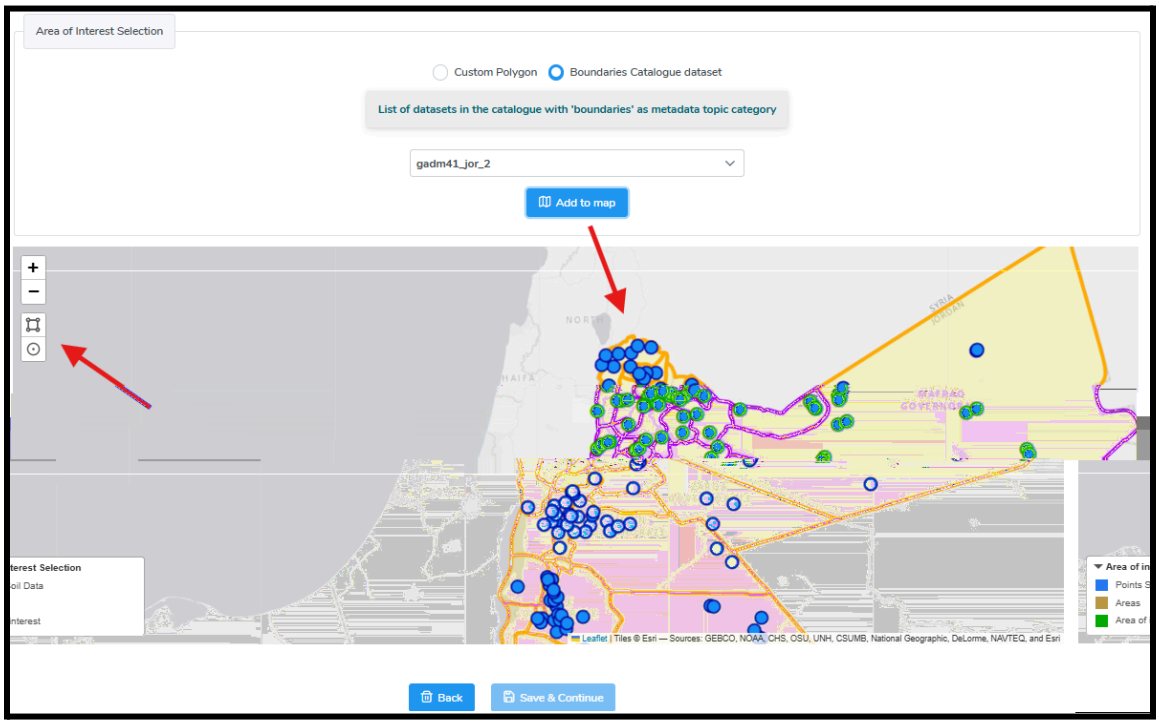


Figure 51 Publishing - Configuration Step 2. Catalogue add source for AOI

After adding the dataset to the map, users can select the required polygons using a point or a selection box, after activating the corresponding tool in the top-left corner of the map.

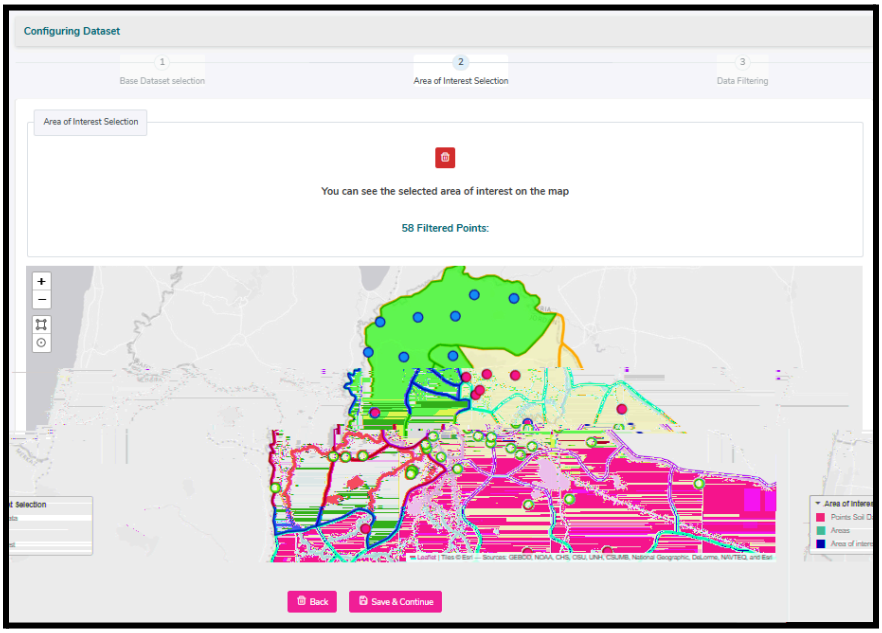


Figure 52 Publishing - Configuration Step 2. AOI selection

After the AOI has been selected, the number of filtered points is displayed on the page. The user can then click “Save & Continue”.

The next configuration step consists of filtering the points according to selected attributes s:

- **observation period** start and end dates, or all dates
- **type :** *Auger hole, Minipit* or *Profile,* or all types
- **project**: the Identifier or name of the project, survey activity/campaign, within which the point based soil data were collected or all projects
- The **depth:** upper and lower depth, where available. User can select the 0- 20 cm interval , the 20 to 50 cm interval , or all depths
- **laboratory method:** the method used for the measurements, no method or all methods. This filter is available depending on the measurements involved and applies only to soil indicators.
- **the field survey** method: the method used to carry out field survey, or all methods.

Identifier	Code	Description
SURVEY_METHODS:CP	CP	method: CPCS;
SURVEY_METHODS:FA	FA	method: FAO
SURVEY_METHODS:FA06	FA06	method: FAO; year: 2006
SURVEY_METHODS:FA52	FA52	method: FAO; year: 1952
SURVEY_METHODS:FA77	FA77	method: FAO; year: 1977
SURVEY_METHODS:FA90	FA90	method: FAO; year: 1990
SURVEY_METHODS:US	US	method: USDA soil survey manual
SURVEY_METHODS:US18	US18	method: USDA soil survey manual ; year: 2018
SURVEY_METHODS:US52	US52	method: USDA soil survey manual ; year: 1952
SURVEY_METHODS:US62	US62	method: USDA soil survey manual ; year: 1962
SURVEY_METHODS:US75	US75	method: USDA soil survey manual ; year: 1975
SURVEY_METHODS:US93	US93	method: USDA soil survey manual ; year: 1993
SURVEY_METHODS:WR	WR	method: WRB;
SURVEY_METHODS:WR22	WR22	method: WRB; year: 2022

Figure 53 Survey Methods

The filtering panel (Figure 54) displays a form for applying the filters and a map for visualizing the result.

The information at the top of the page indicates the number of selected points selected and the number of aggregated points required for interpolation. The aggregation is based on the location of the points, using a10m point buffer and on a weighted average of values based on depth intervals (for soil indicators only). The values available in the drop down lists correspond to the data contained in the selected points.

After applying the filters, the user can click “**Save & Continue**”.

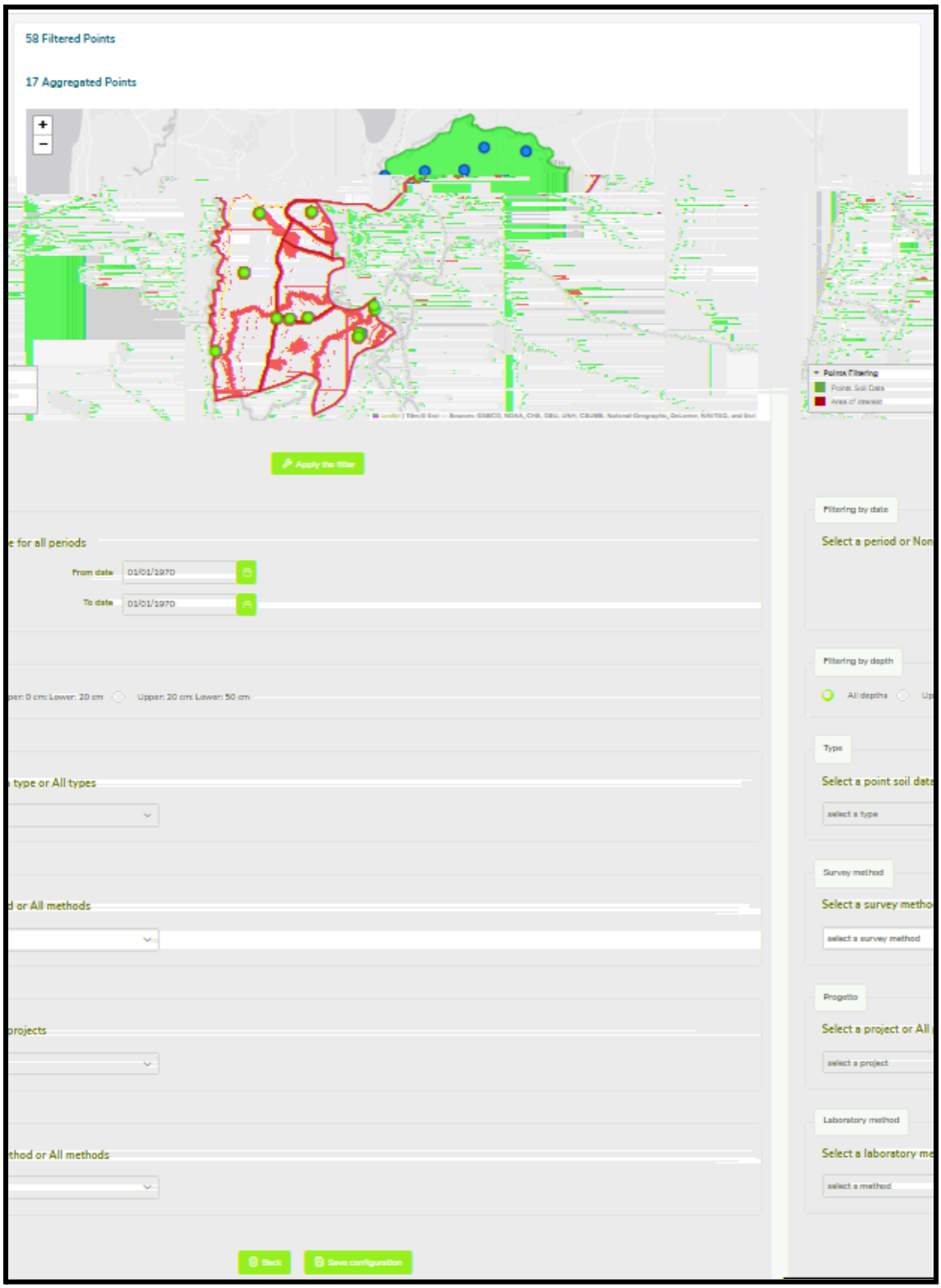


Figure 54 Publishing - Configuration - Filtering panel

The customized dataset then enters the validation phase.

The system displays a report showing the number of filtered and aggregated points and indicating whether the dataset contains **outliers**.

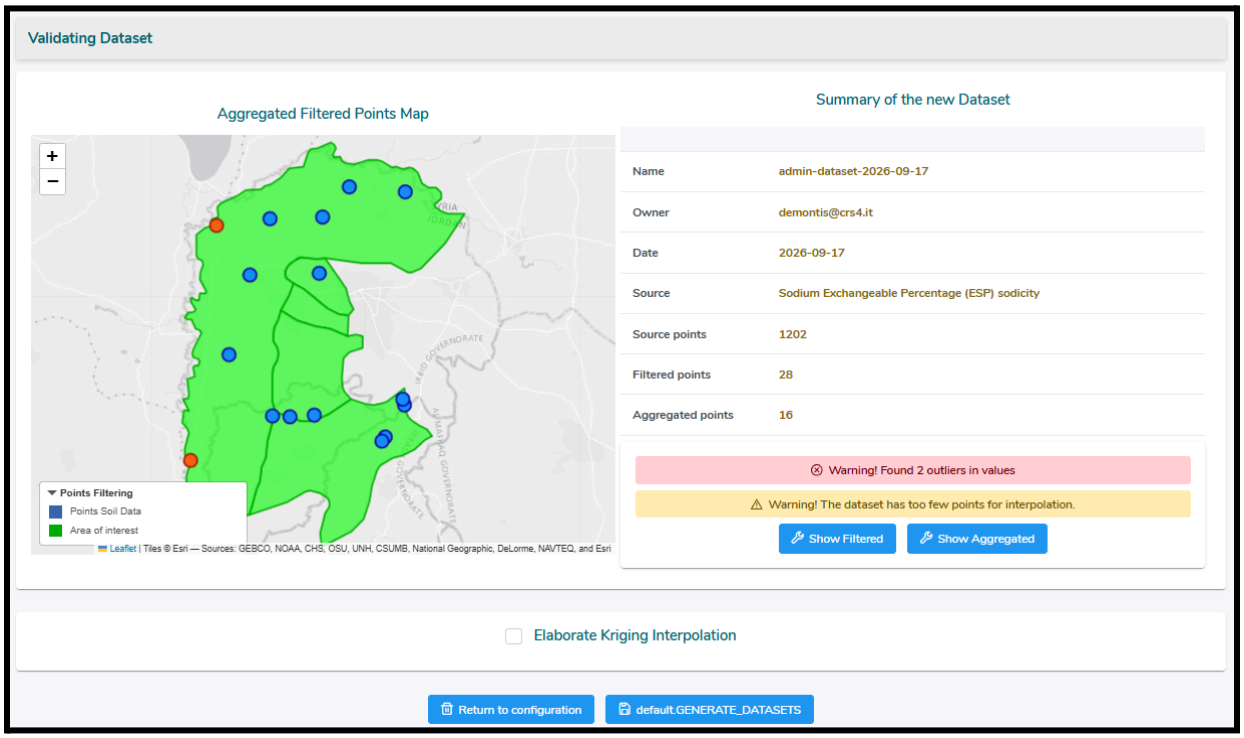


Figure 55 Publishing - Validation - Report panel

The presence of outliers may result in predictions outside the expected range by skewing parameter estimates and **altering the underlying data distribution**.

Outliers can result from data entry errors, but they can also represent genuine data points that WARRANT FURTHER INVESTIGATION.

If the dataset does not require further processing, such as Kriging interpolation, it can be published directly in the catalogue.

Otherwise, selecting the "Elaborate Kriging Interpolation" checkbox opens the Kriging interpolation panel (Figure 56).

In this panel, users can set the interpolation parameters and evaluate the resulting semivariogram (Figure 57)

If the validation results are satisfactory, the data manager can proceed with the generation of the new datasets.

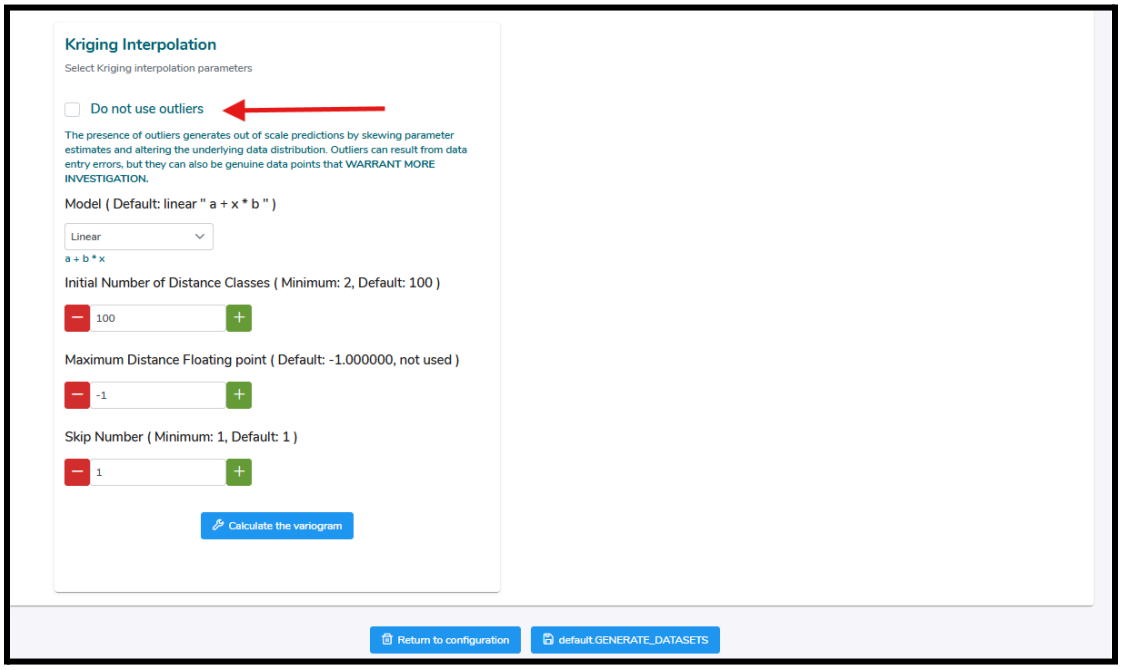


Figure 56 Publishing - Validation - Kriging Interpolation

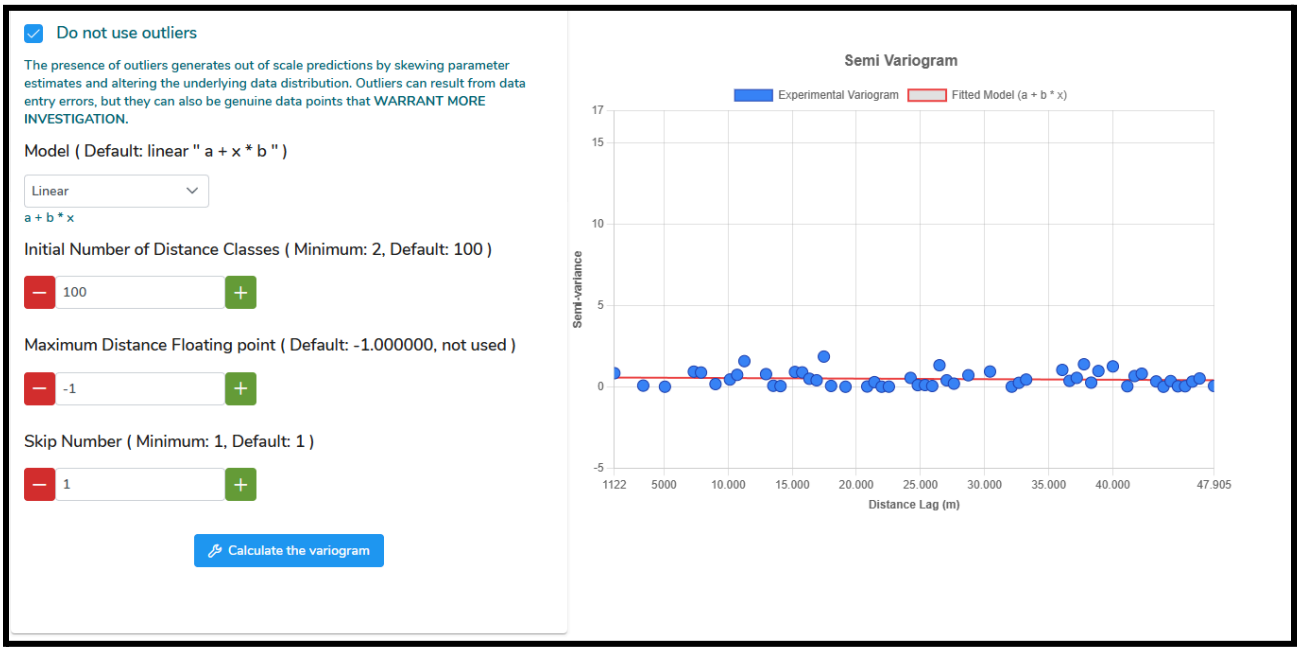


Figure 57 Publishing - Validation - Kriging Interpolation - Semi variogram

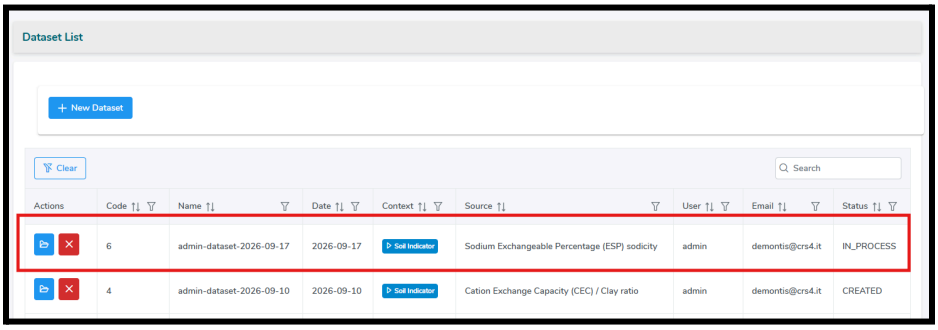


Figure 58 Publishing - Datasets in elaboration

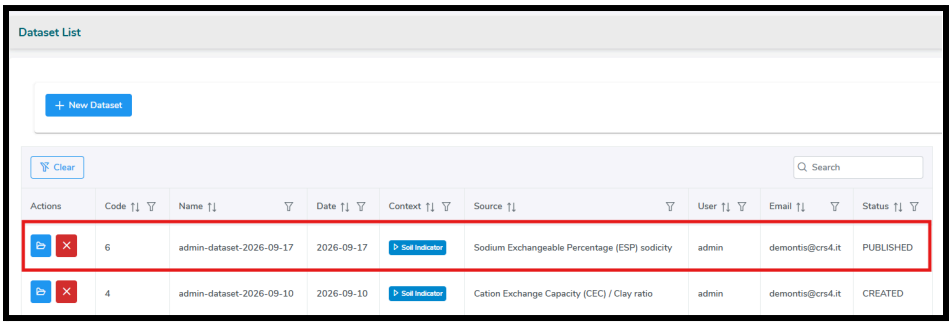


Figure 59 Publishing - Datasets published

After processing, the customized dataset is published in the catalogue. The publishing report proposes a style (Figure 60) for the raster dataset because, since the generated raster dataset is assigned a default style



Figure 60 Publishing - Report

To change the default raster style, open the dataset in the catalogue and select “Edit” menu then “Edit style”

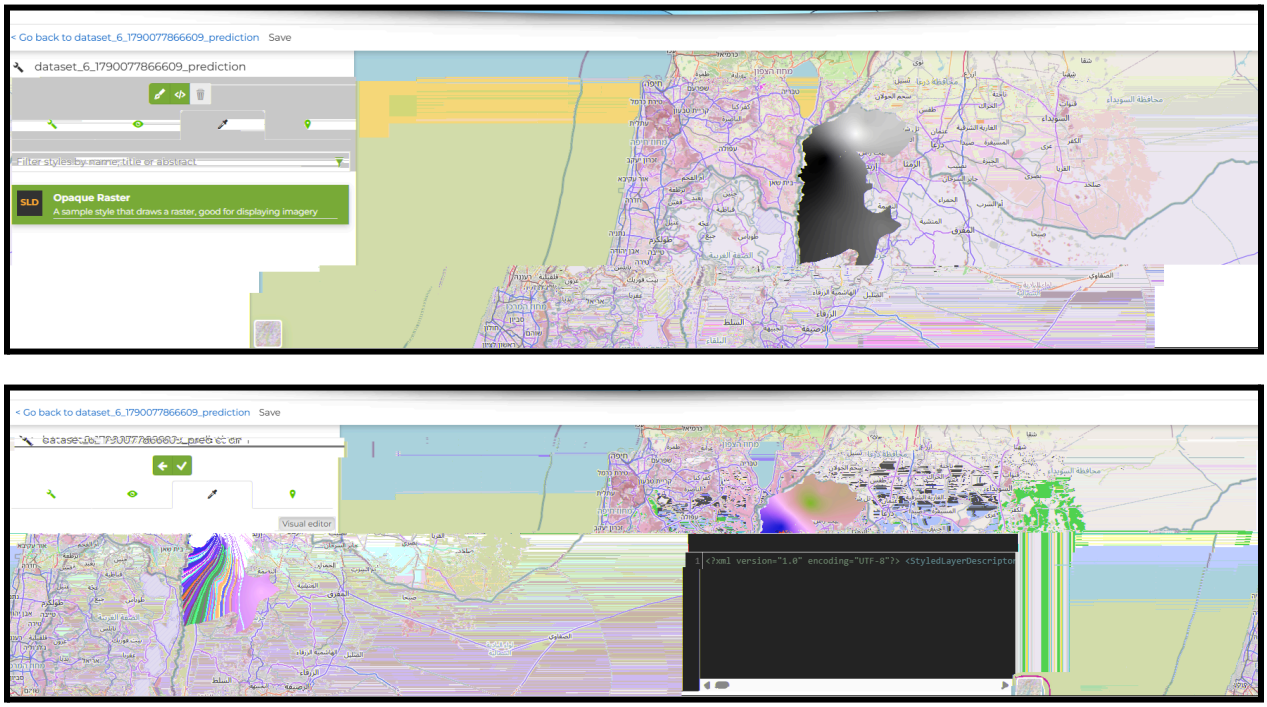


Figure 61 Publishing - changing raster style

Finally, data managers **must edit the metadata and change the name** of the generated datasets in the catalogue. They must then assign the appropriate permissions to the End User allowing them to view and download the datasets.

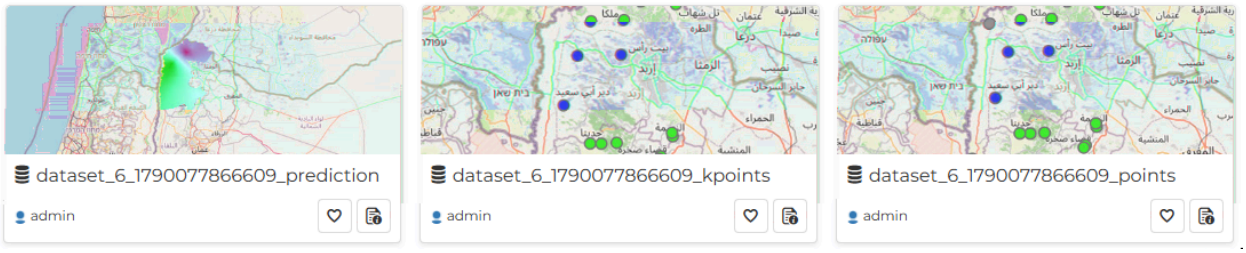


Figure 62 Publishing - generated rasters

The customized dataset generated by the publishing tool are:

- raster dataset containing the interpolation results (optional)
- point vector dataset of aggregated points, whose name ends with '_kpoints' This dataset can be used to perform further interpolation in desktop GIS software.
- point vector dataset of filtered points, whose name ends with '_points'.

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4. API

The MED-SIS uses API managing Catalogue resources and Soil Data

The Geonode API is described in several web document such as
https://docs.geonode.org/projects/v4/tml

5. Troubleshooting

- **The data and legend of a dataset is not displayed in views.**
  The legend's name shows a "!"

  The problem may stem from:

  1) A problem with the dataset's style (SLD code)
     -> Go to the Style configuration and verify that is ok or not

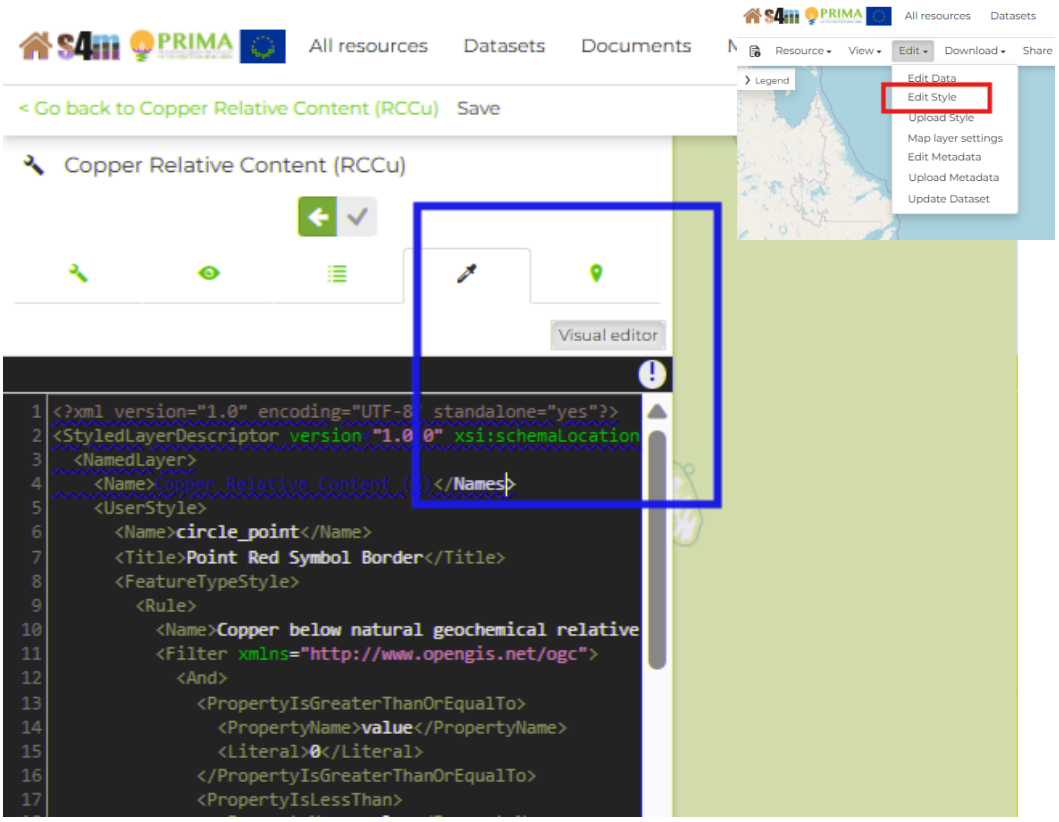


Figure 64

  2) The communication between GeoNode and GeoServer regarding dataset permissions for groups.
     -> If the user is a "Data Manager," they can modify the dataset permissions by assigning themselves management permissions. If the user is not a "Data Manager," the layer permissions must be modified to grant the user "download" permission.

3) Outdated browser data when switching users;
-> specifically, logging out as User A and then logging in as User B may result in User A remaining logged into GeoServer. To resolve the issue, open GeoServer and perform a logout.

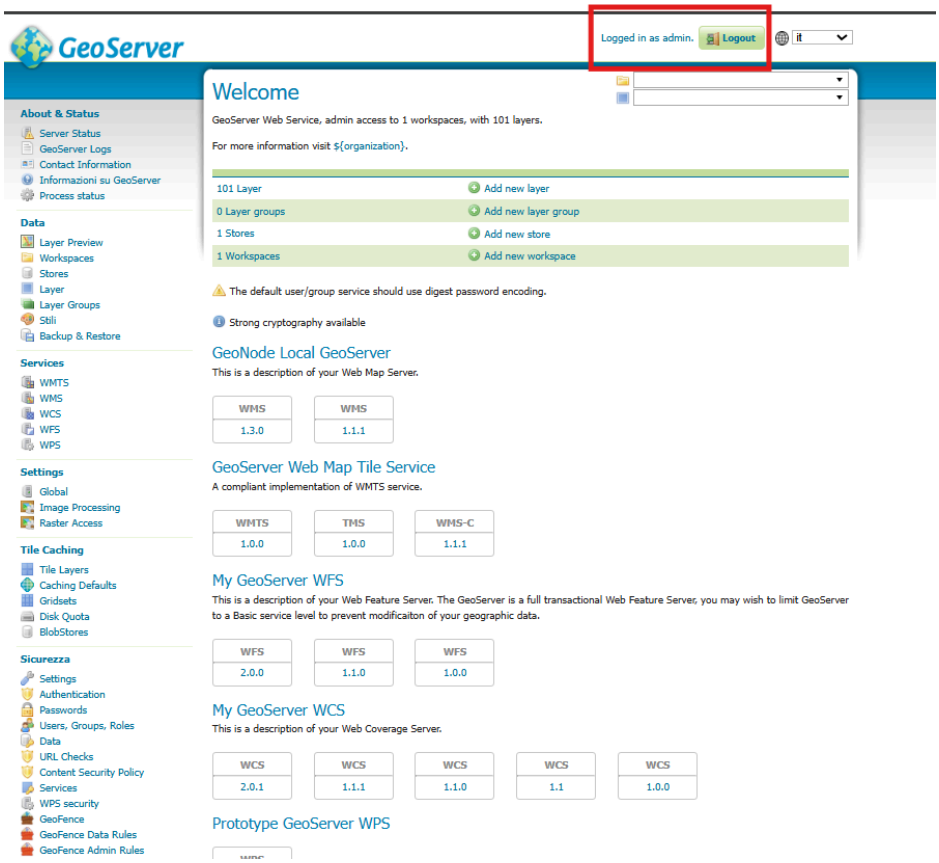


Figure 65: Geoserver log out